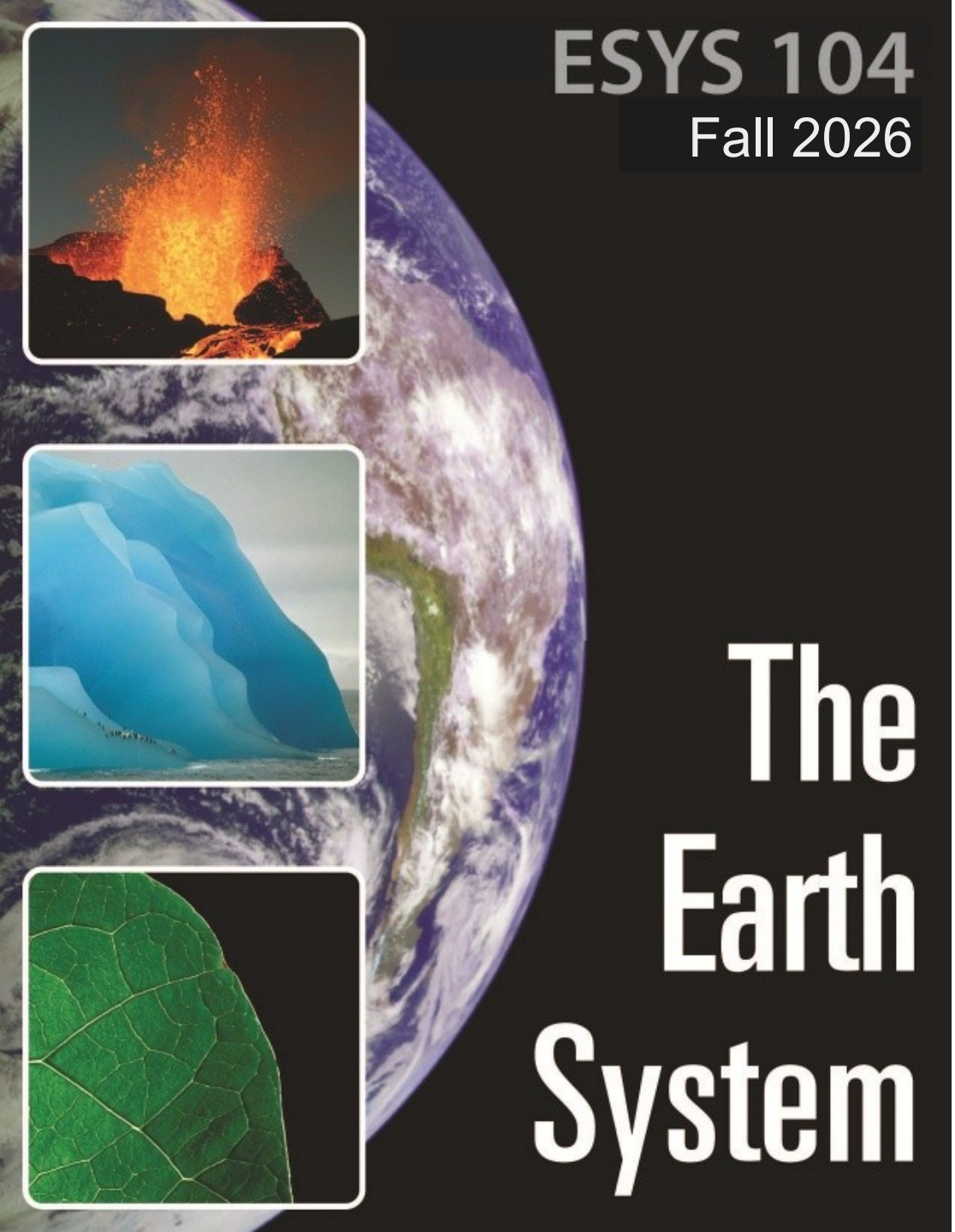


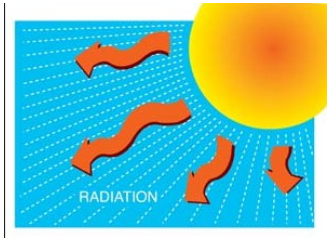
#5: Atmosphere energy balance

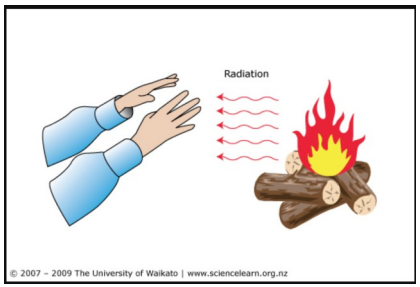
Quiz #1 today
Assignment #1 posted later today
(both on myCourses)

Energy Budget of the Earth

Sensible Heat

Radiation

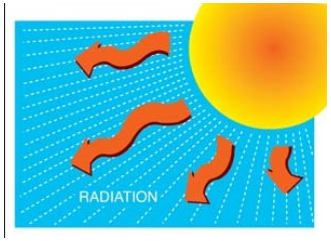


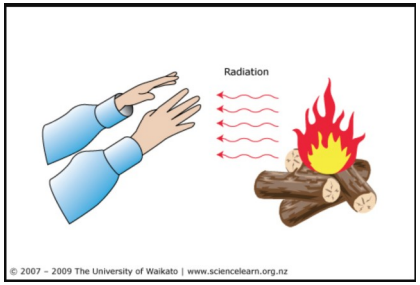


Energy Budget of the Earth

Sensible Heat

Radiation

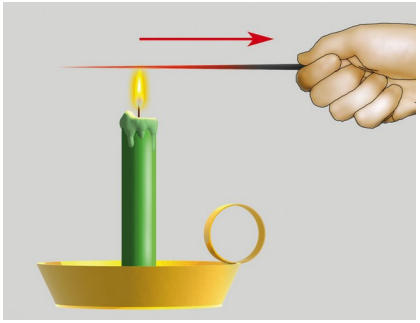






Convection

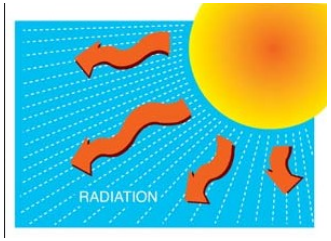
Conduction

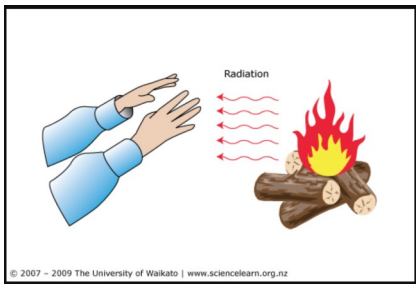


Energy Budget of the Earth

Sensible Heat

Radiation

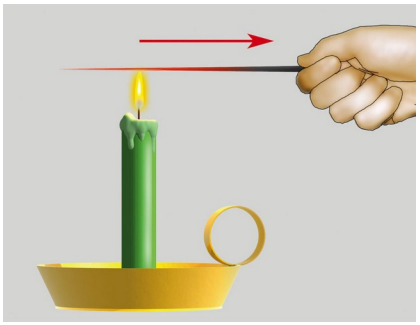






Convection

Conduction



Latent Heat



The Atmosphere



Composition of the Atmosphere

	% by volume
Nitrogen (N_2)	78%
Oxygen (O_2)	21%
Argon (Ar)	1%
Water Vapor (H_2O)	0 to 4%
Carbon dioxide (CO_2)	0.04%
Methane (CH_4)	0.0002%

Earth's Atmosphere Composition

PERMANENT GASES			VARIABLE GASES			
Gas	*Symbol*	*Percent (by Volume) Dry Air*	*Gas (and Particles)*	*Symbol*	*Percent (by Volume)*	*Parts per Million (ppm)**
Nitrogen	N_2	78.08	Water vapor	H_2O	0 to 4	
Oxygen	O_2	20.95	Carbon dioxide	CO_2	0.038	380*
Argon	Ar	0.93	Methane	CH_4	0.00017	1.7
Neon	Ne	0.0018	Nitrous oxide	N_2O	0.00003	0.3
Helium3	He	0.0005	Ozone	O_3	0.000004	0.04**
Hydrogen	H_2	0.00006	Particles (dust, soot, etc.)		0.000001	0.01–0.15
Xenon	Xe	0.000009	Chlorofluorocarbons (CFCs)		0.00000002	0.0002

*For CO_2, 380 parts per million means that out of every million air molecules, 380 are CO_2 molecules.

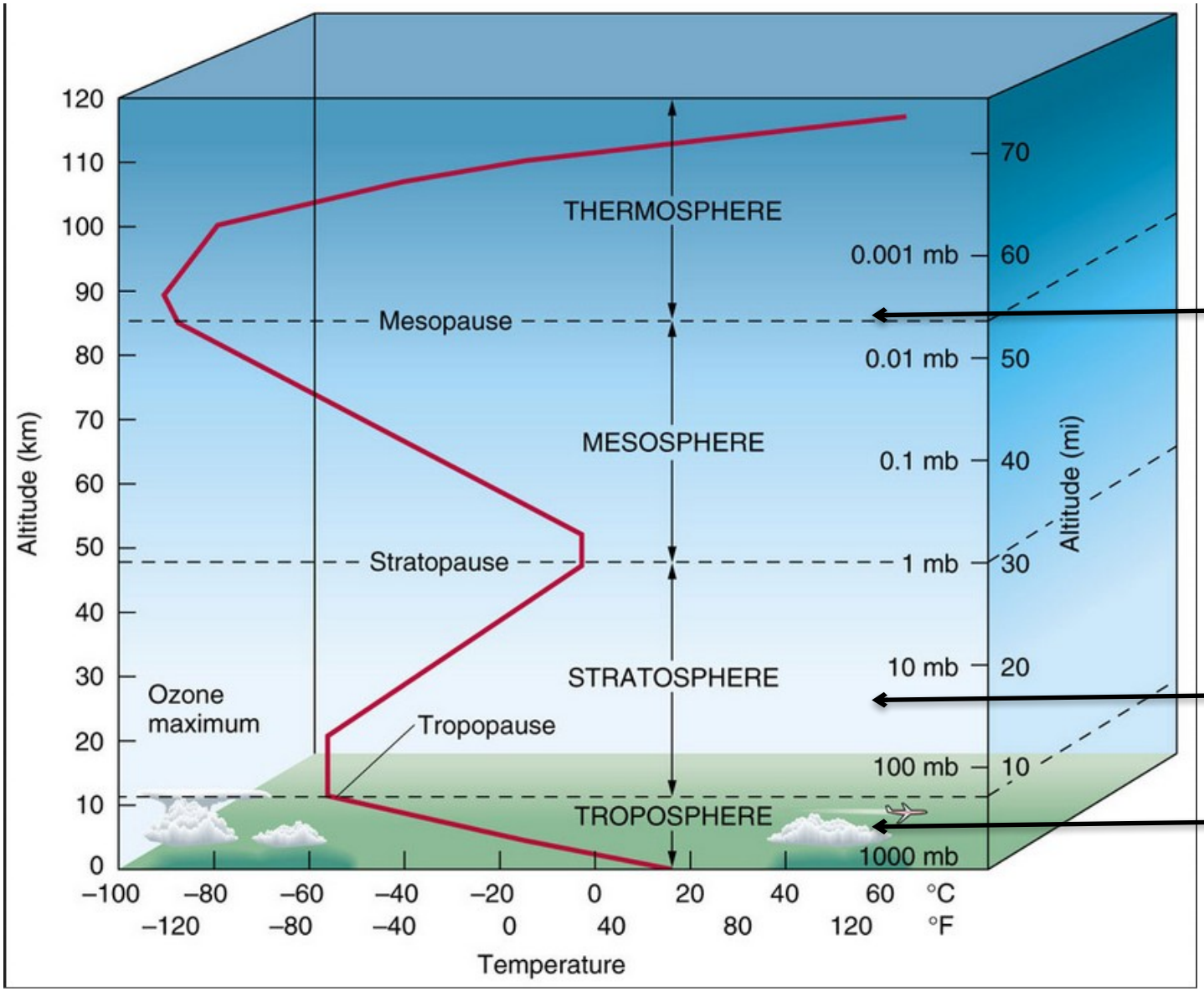
**Stratospheric values at altitudes between 11 km and 50 km are about 5 to 12 ppm.

© 2007 Thomson Higher Education

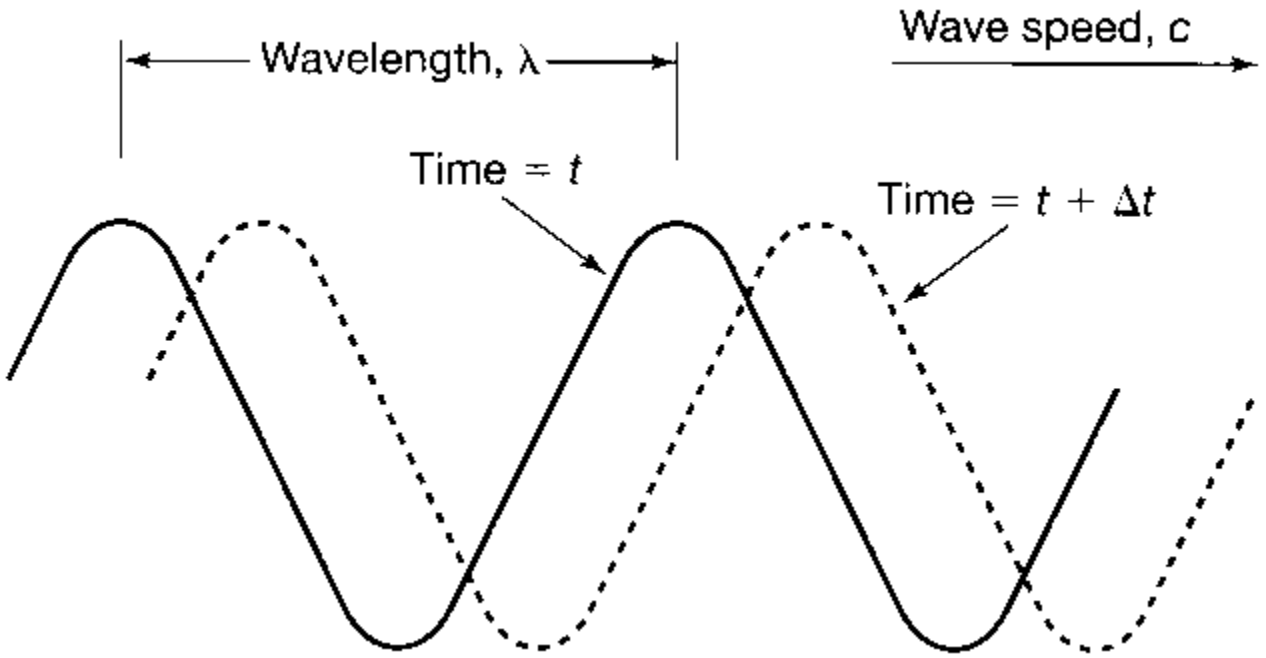
Percent and Parts per Million (ppm) both express fractional compositions

In 2026: CO_2 is now 426 ppm

Vertical Structure of the Atmosphere



Electromagnetic Radiation



All objects emit electromagnetic radiation as waves that are defined by their wavelength and/or frequency.

Electromagnetic Radiation

lambda

λ: wavelength (m/cycle)

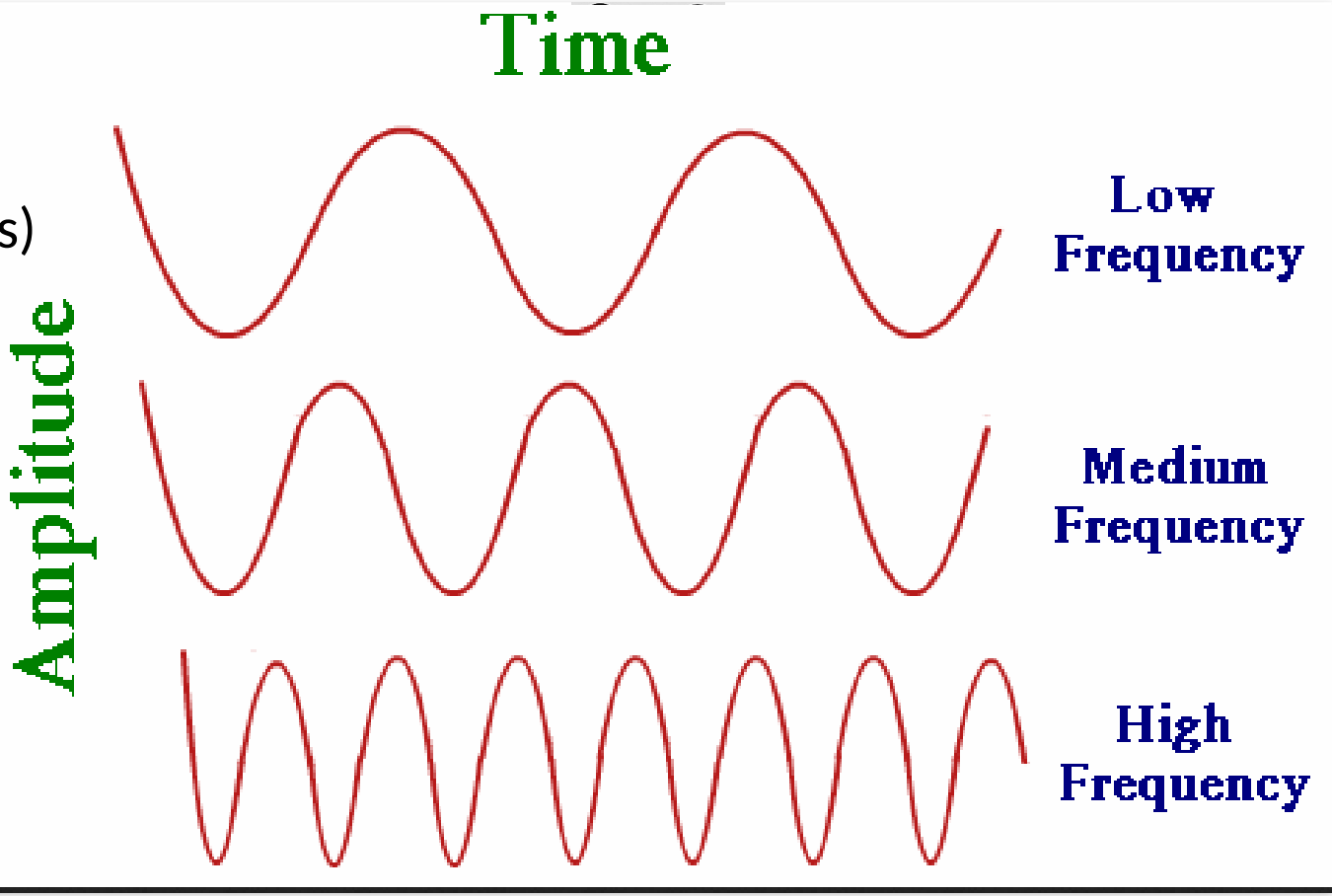
dt: time (s/cycle)

c: speed (of light)

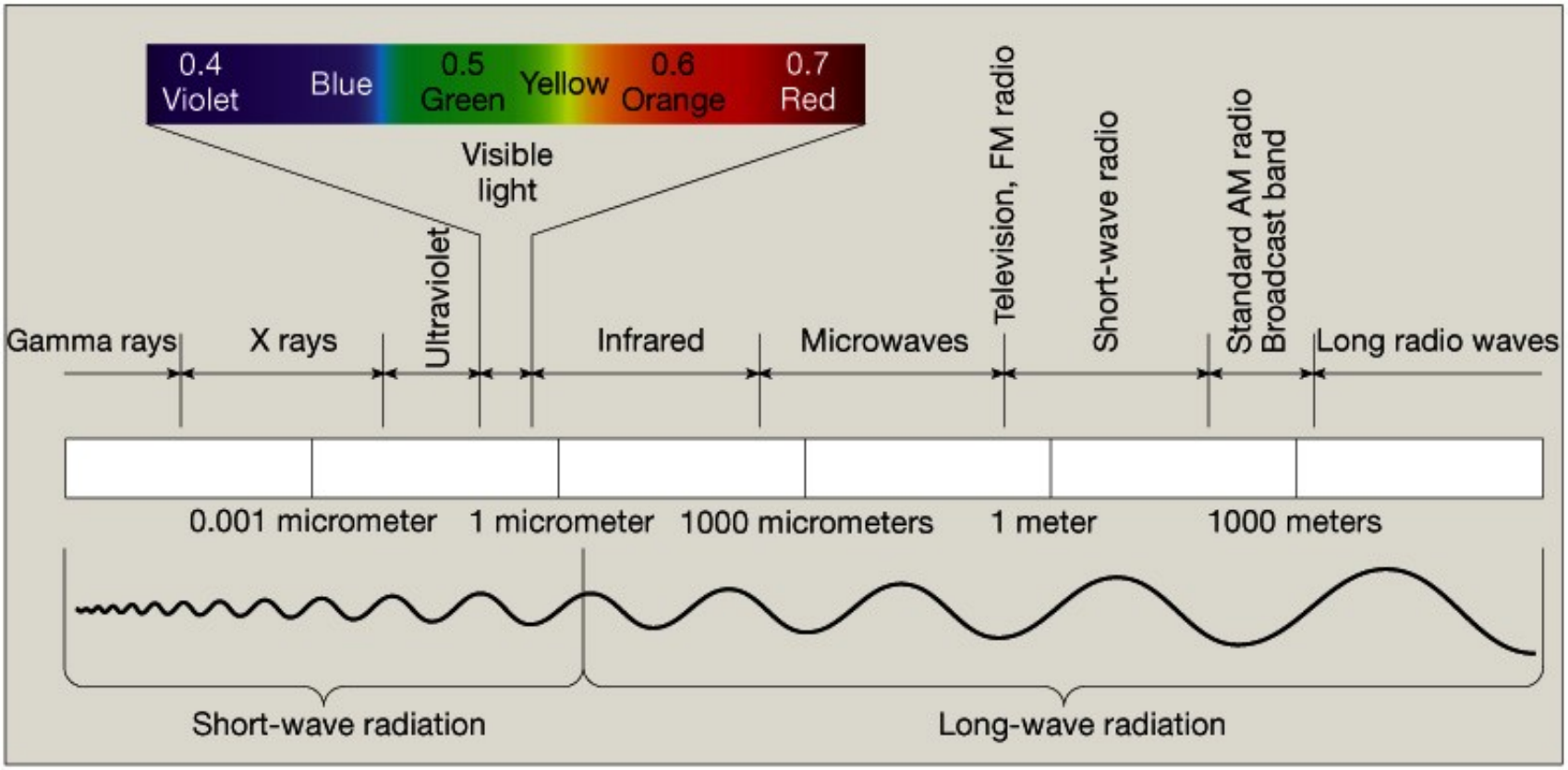
f: frequency (cycle/s)

$c = \lambda / dt$

$c = \lambda f$

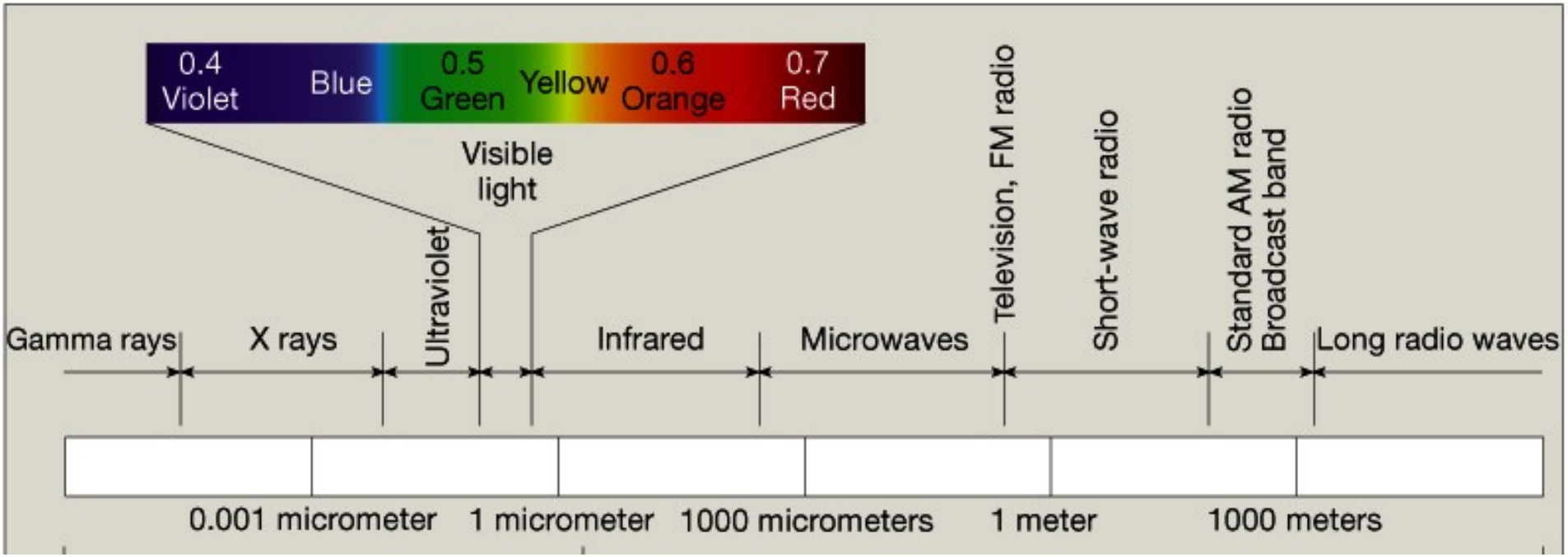


Electromagnetic Radiation



Though the electromagnetic spectrum is continuous, we distinguish different types of waves based on wavelength.

Electromagnetic Radiation

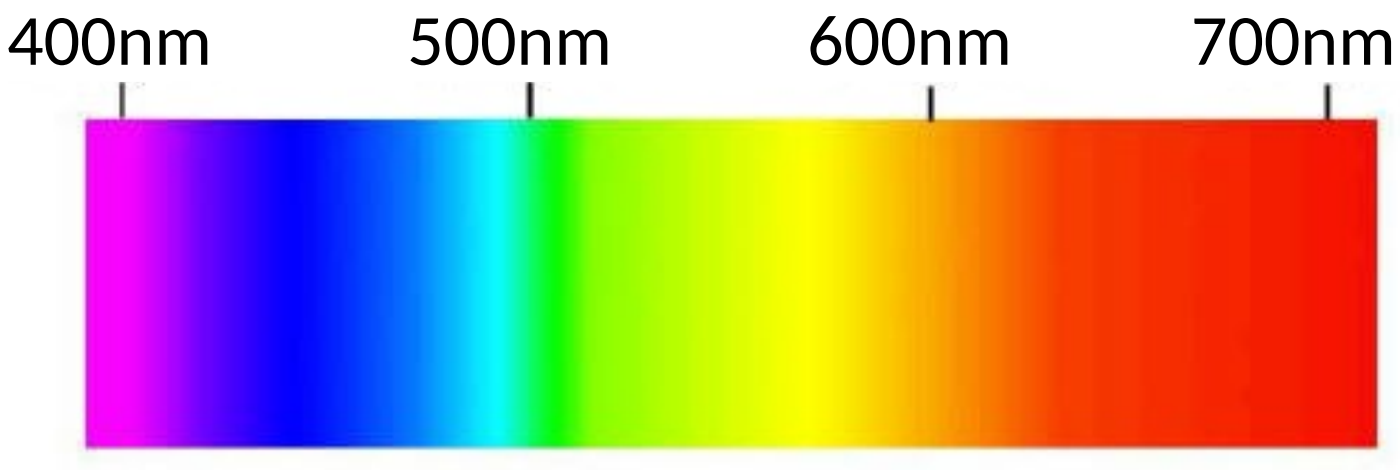


ultraviolet: 0.01 – 0.4 micron

Visible: 0.4 – 0.7 micron

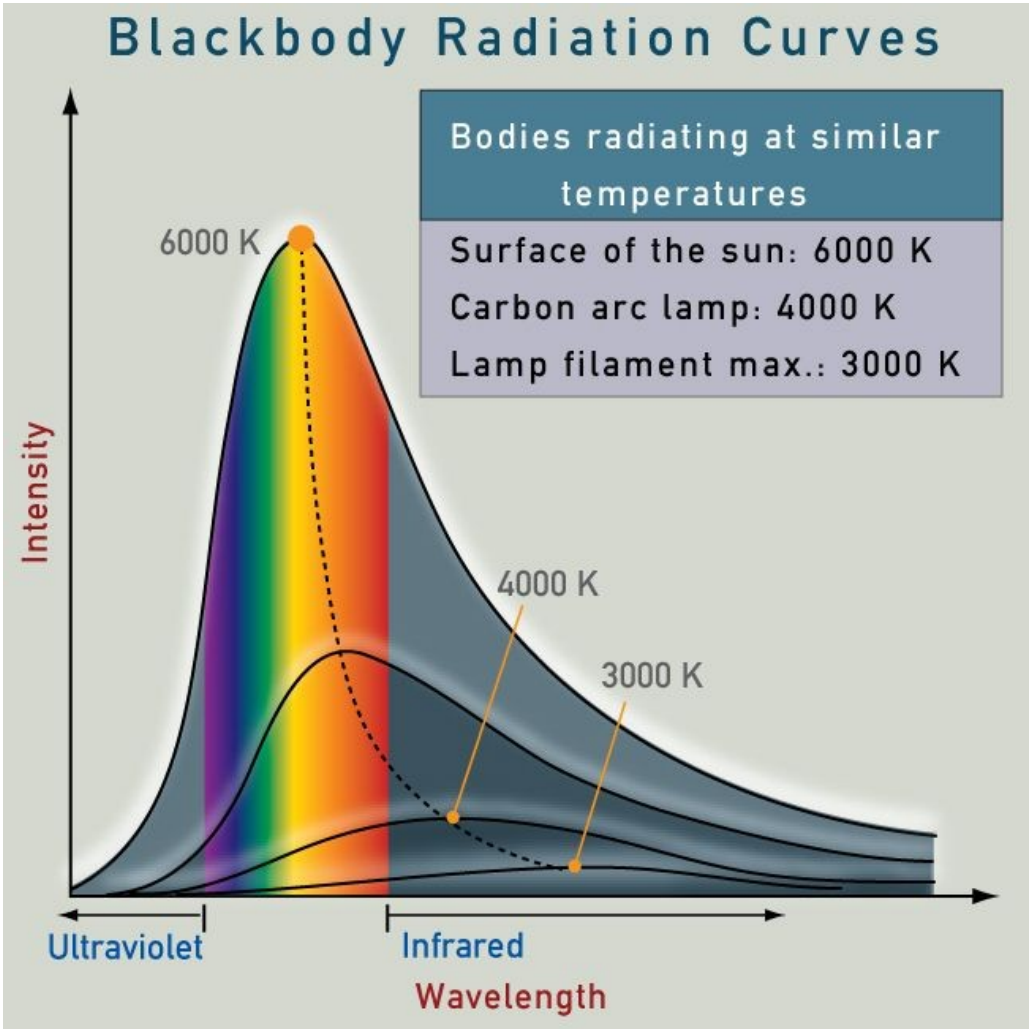
Infrared: 0.7 – 1000 micron

Colors



Reflective colors are made by subtraction (absorption) of some of the colors (some of the wavelengths).

Stefan-Boltzmann Law



$E = eps * sigma * Temp^4$

$E = \epsilon * \sigma * T^4$

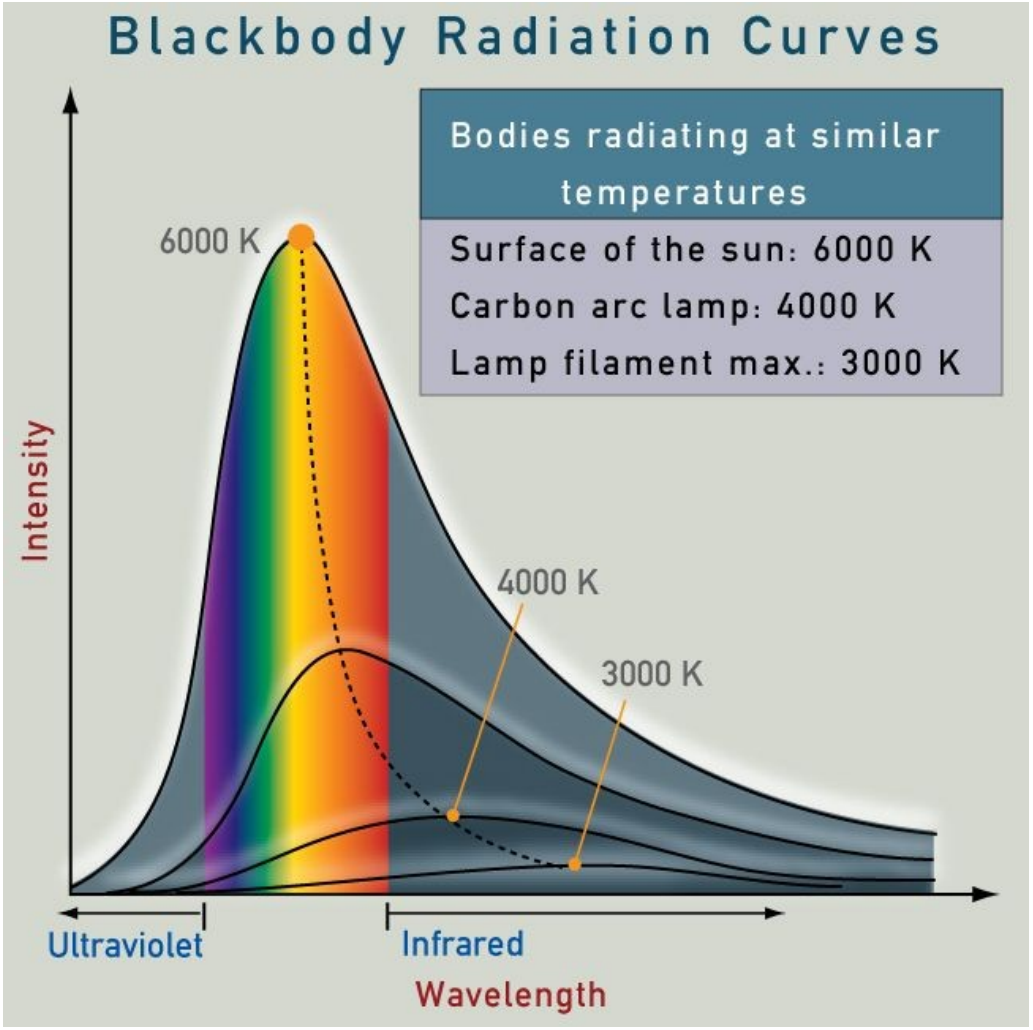
E : Total Energy emitted by a body of temperature T

sigma σ :
the Stefan-Boltzman constant

eps ϵ : emissivity (0 to 1)

a blackbody has an emissivity of one

Wien's Law

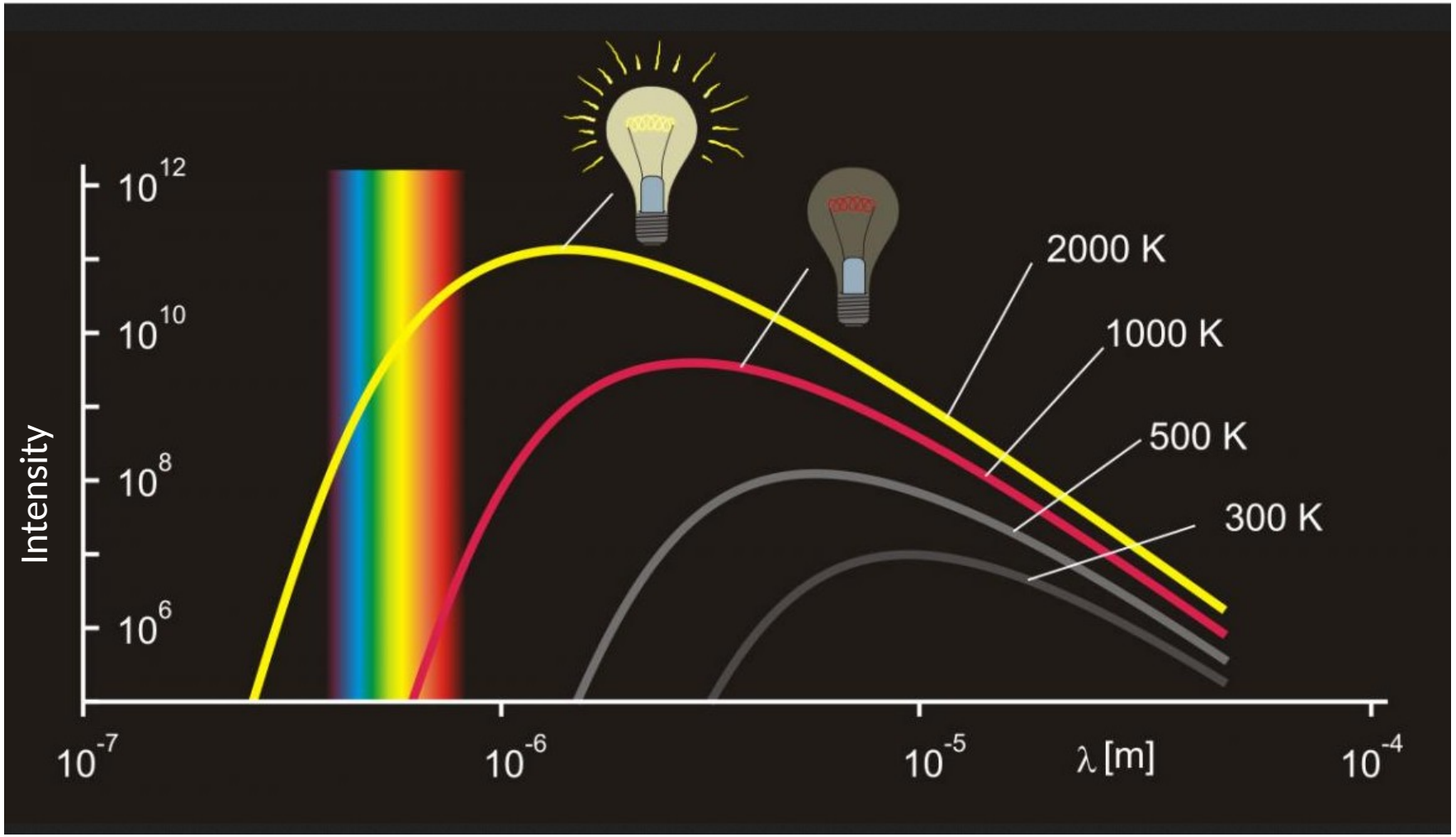


$\lambda_{max} = 2900/T$

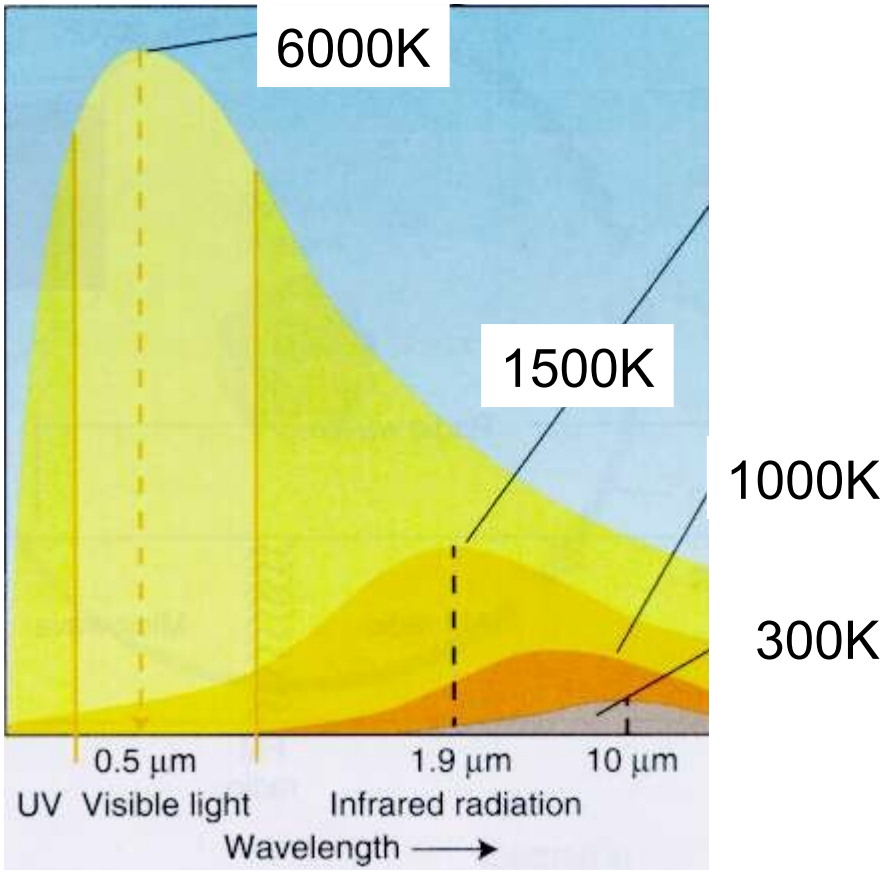
$Lambda_{max} = B / T$

B = constant
= 2900 micron·K
(more precisely)
= 2898 micron·K

Stefan-Boltzmann Law

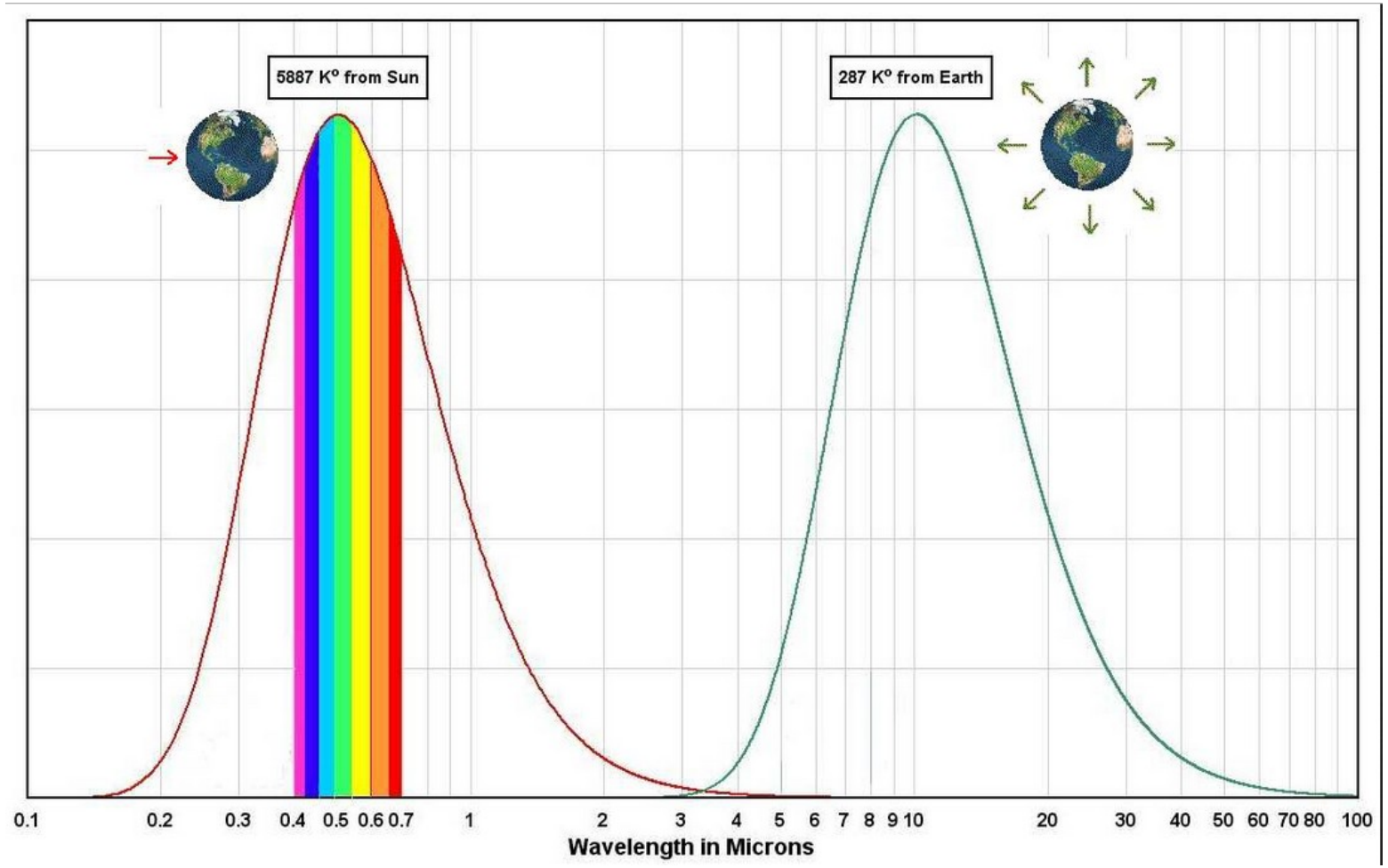


Wavelengths emitted depend on temperature

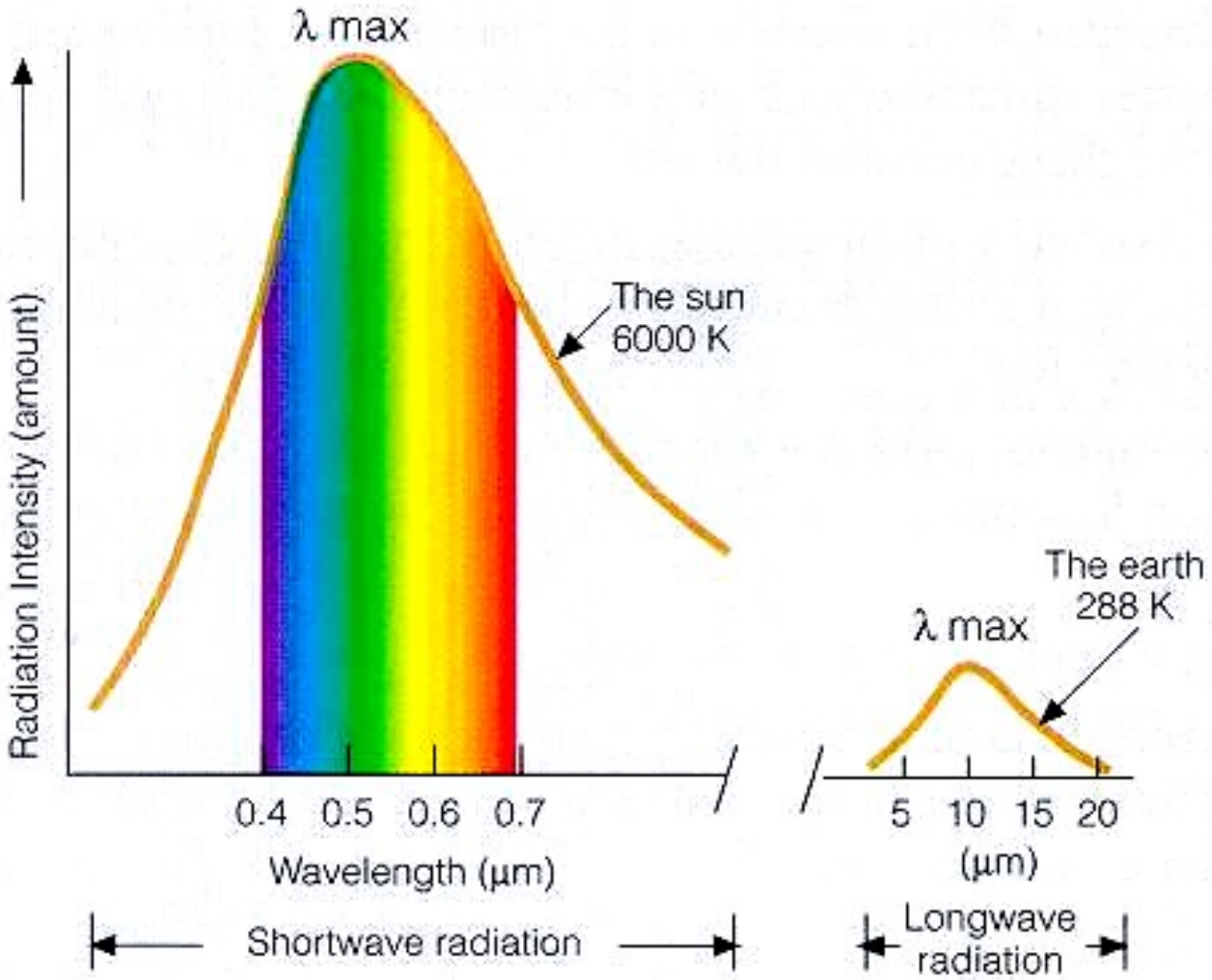


The hotter the object, the shorter the wavelength emitted

Stefan-Boltzmann Law



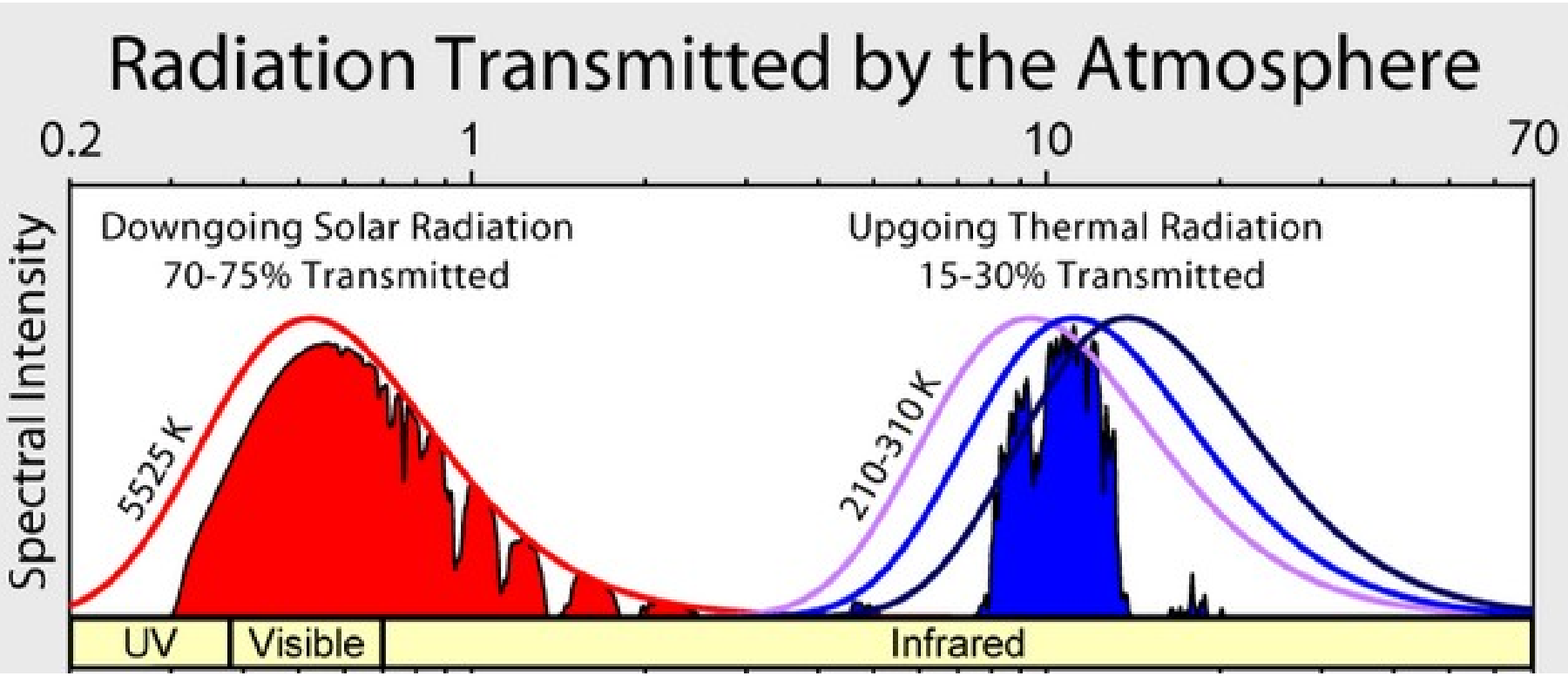
Radiation Properties: Sun vs. Earth

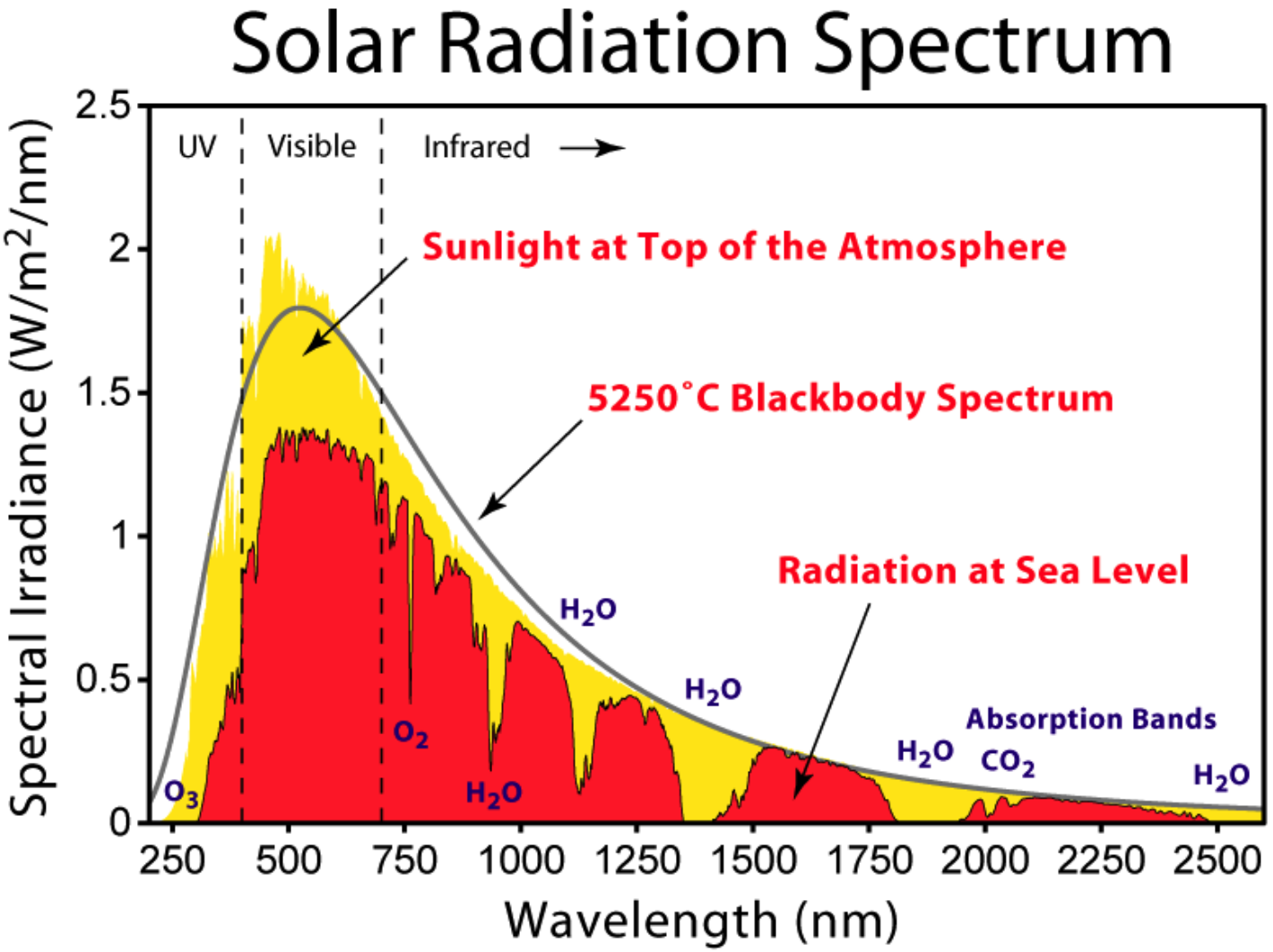


$T_{sun} \approx 5780$ K
→ Sun's peak emission is green

$T_{earth} \approx 288$ K
→ Earth's peak emission is in the infrared

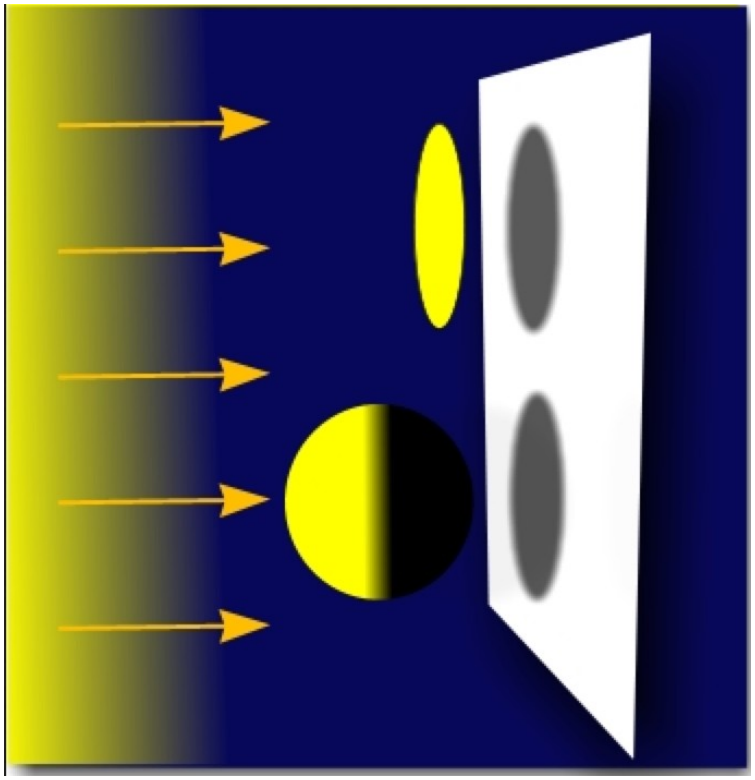
Stefan-Boltzmann Law

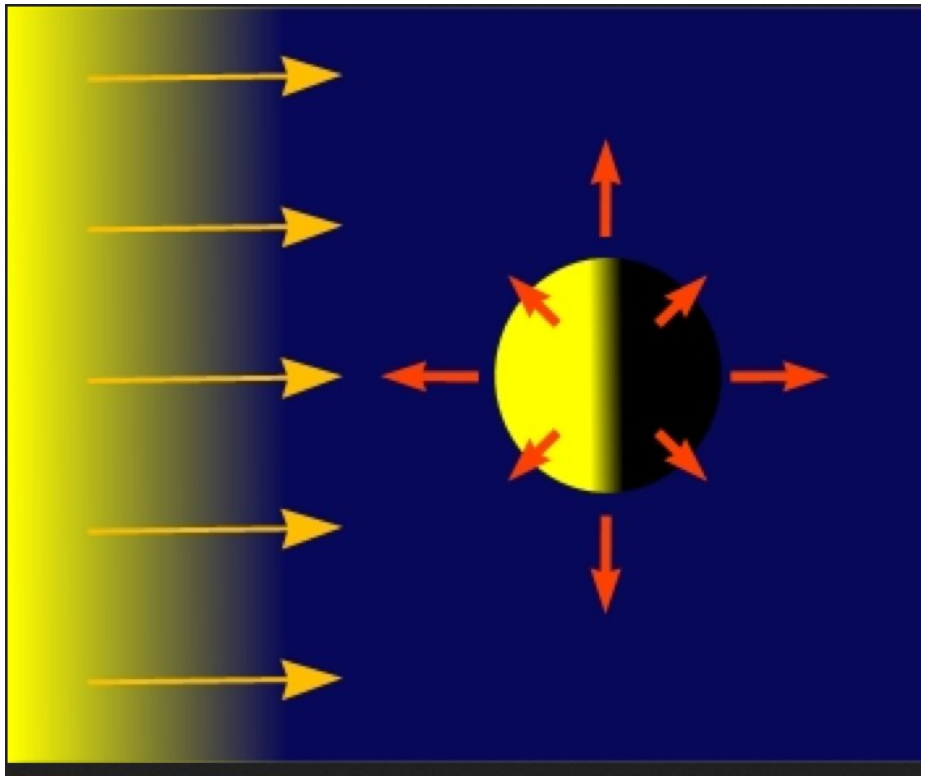



Solar Radiation Spectrum
Spectral Irradiance (W/m²/nm)
Wavelength (nm)
UV
Visible
Infrared
Sunlight at Top of the Atmosphere
5250˚C Blackbody Spectrum
Radiation at Sea Level
Absorption Bands
O3
O2
H2O
H2O
H2O
H2O
CO2
H2O
0
0.5
1
1.5
2
2.5
250
500
750
1000
1250
1500
1750
2000
2250
2500

Radiative Balance

$S_0 = 1368\ W/m^2$





$E_{in} = ??$

$E_{out} = ??$

Radiative Balance Terms

Solar constant at Earth's orbit S_0 = 1368 W/m^2

Albedo (reflectivity)

Atmospheric absorption

short wave: ozone, particulates

long wave: H_2O, CO_2, CH_4, etc.

Energy emitted: blackbody $E_{out} = \sigma T_E^4$

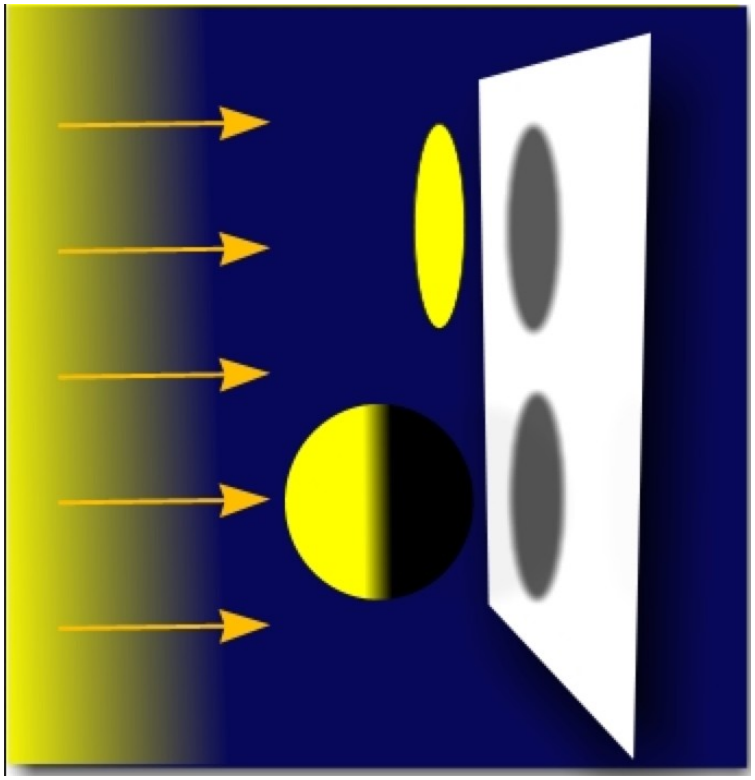
graybody $E_{out} = \epsilon \sigma T_E^4$

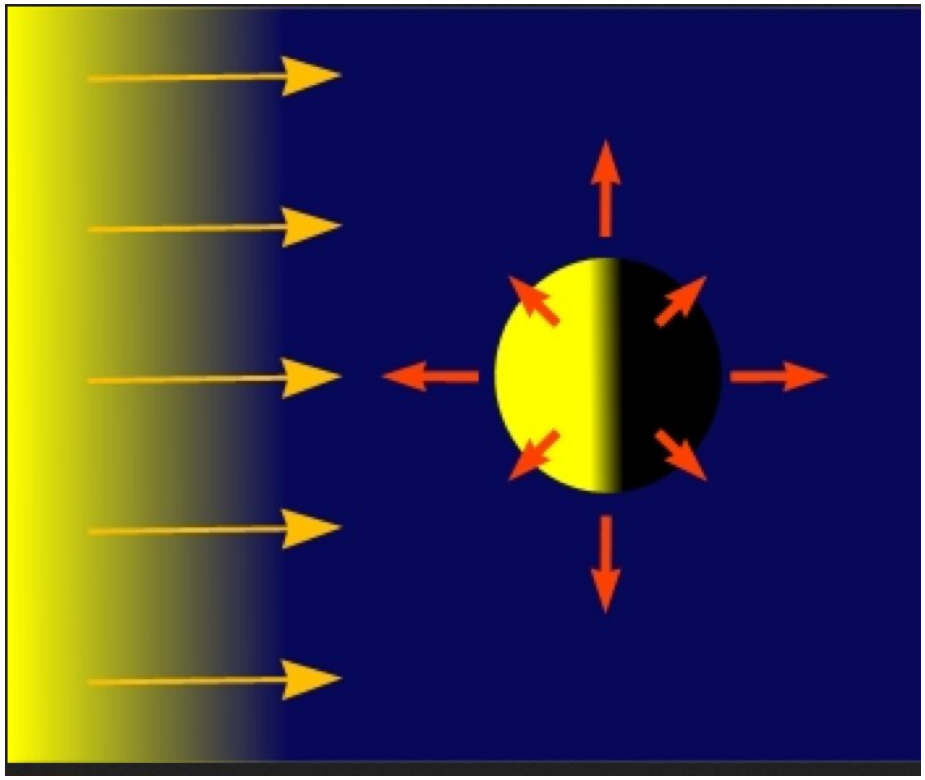
Stefan-Boltzman constant sigma σ = 5.67 x 10^{-8} W/(m^{2}K^4)

Emissivity (eps) ϵ from 0 to 1 (one is a perfect black body)

Radiative Balance

$S_0 = 1368\ W/m^2$

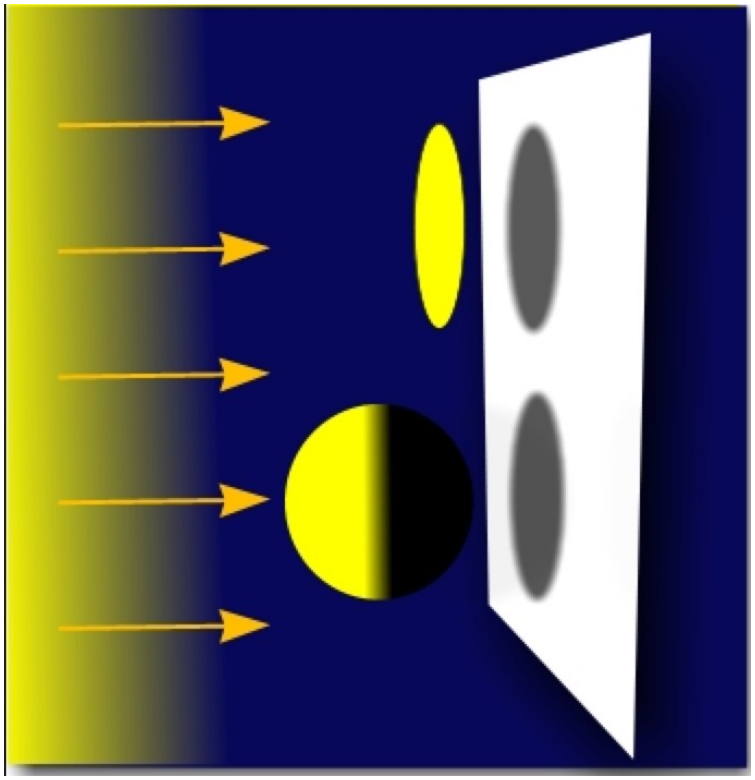


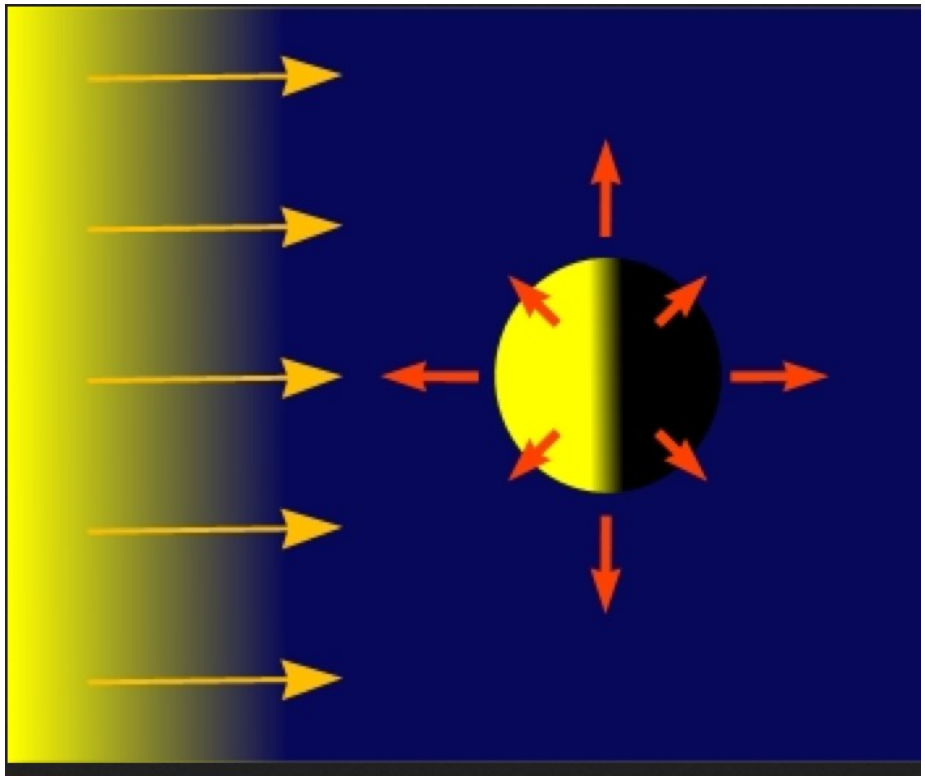


$$E_{in} = S_0\ \pi\ R_E^2 \quad = \quad E_{out} = \sigma\ T_E^4\ 4\ \pi\ R_E^2$$

Radiative Balance

$S_0 = 1368\ W/m^2$

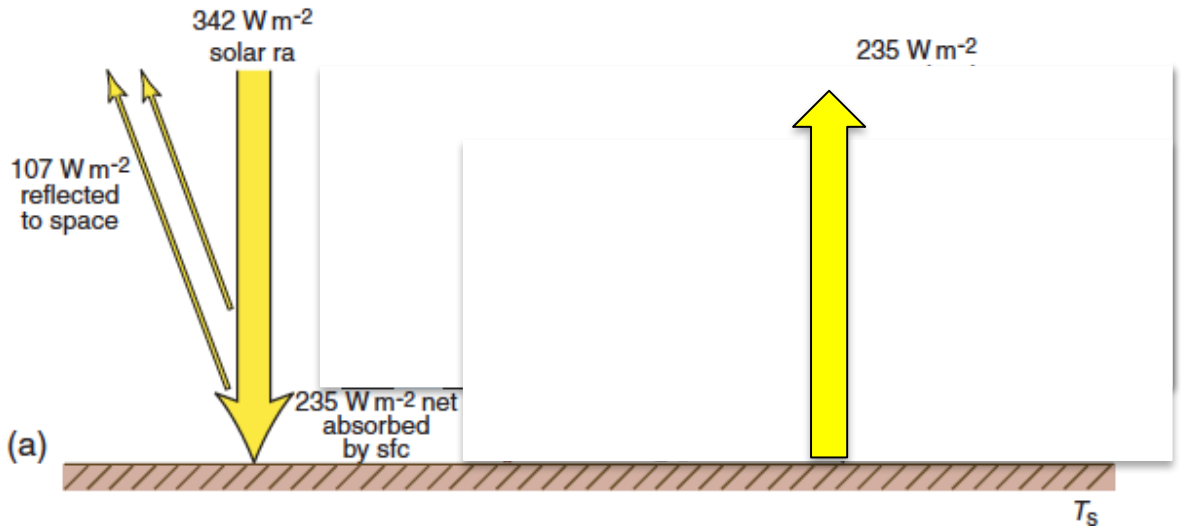




$$T_E = (S_0 / (4\ \sigma))^{1/4}$$

Sigma $\sigma = 5.67 \times 10^{-8}\ W/(m^2K^4)$

Energy Balance

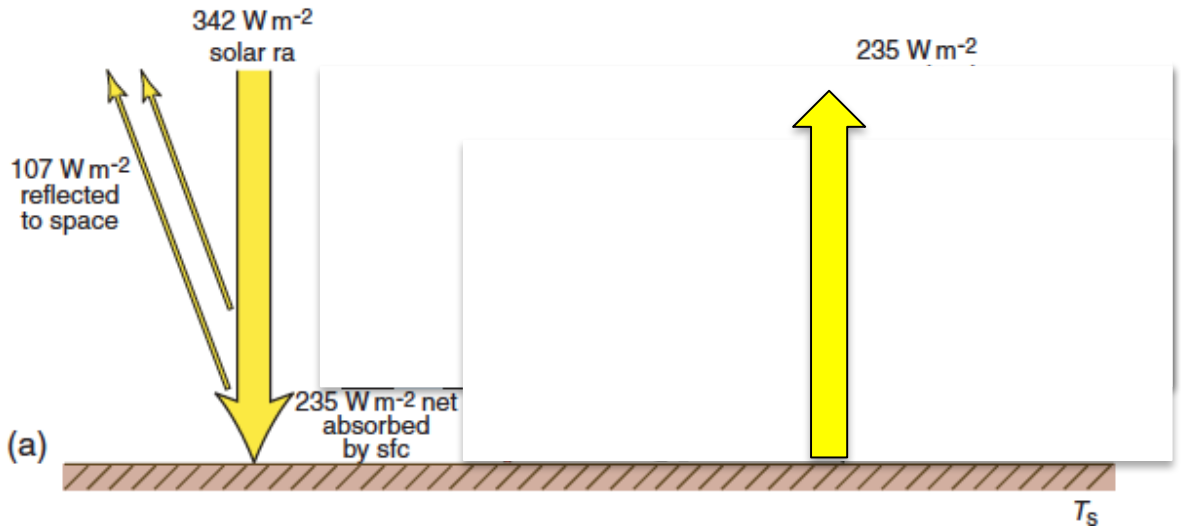


E_{input} = ?? = E_{output} = ??

What is the Planetary Albedo (α alpha)?

What is the Radiative Equilibrium Temperature?

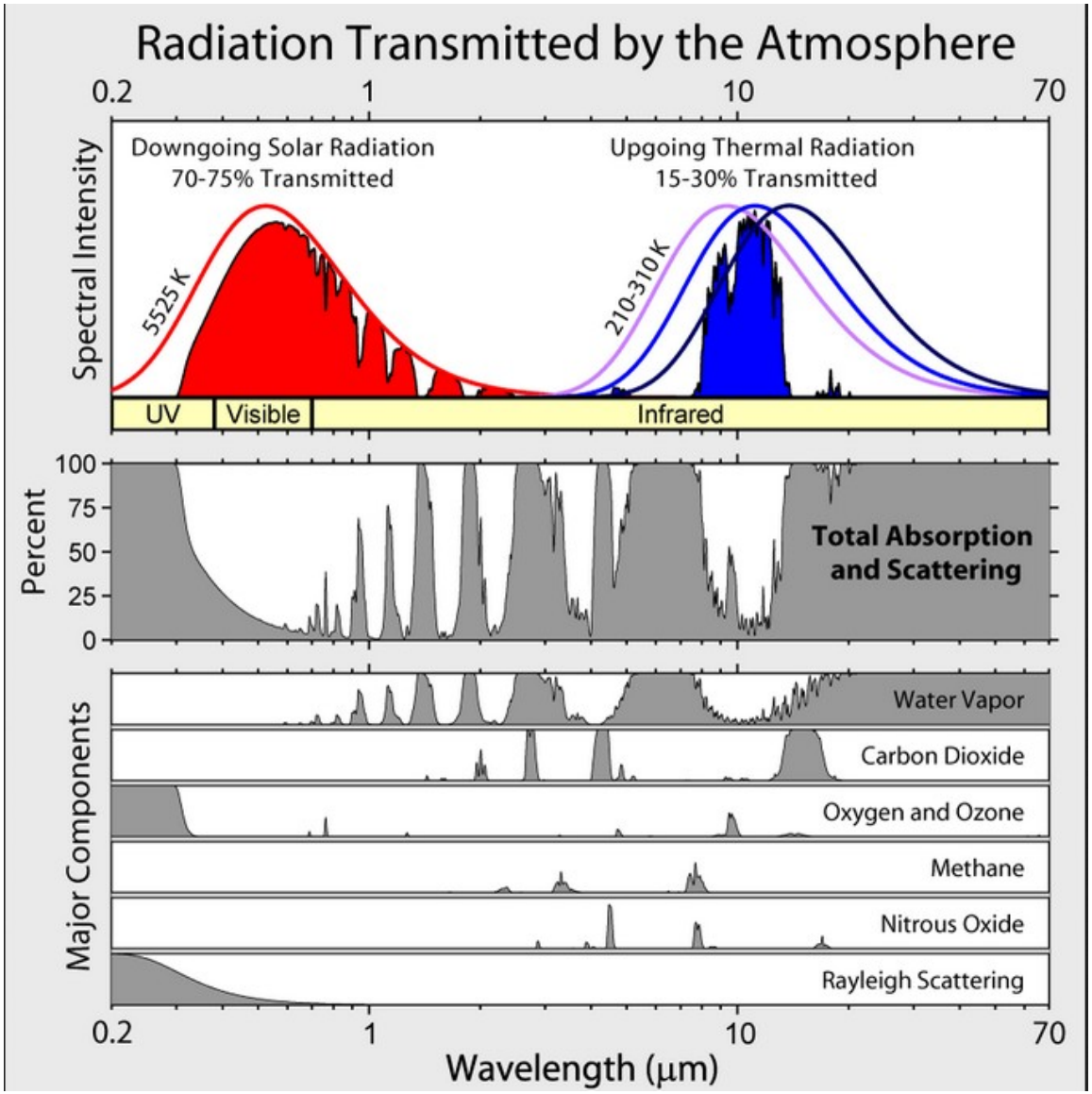
Energy Balance



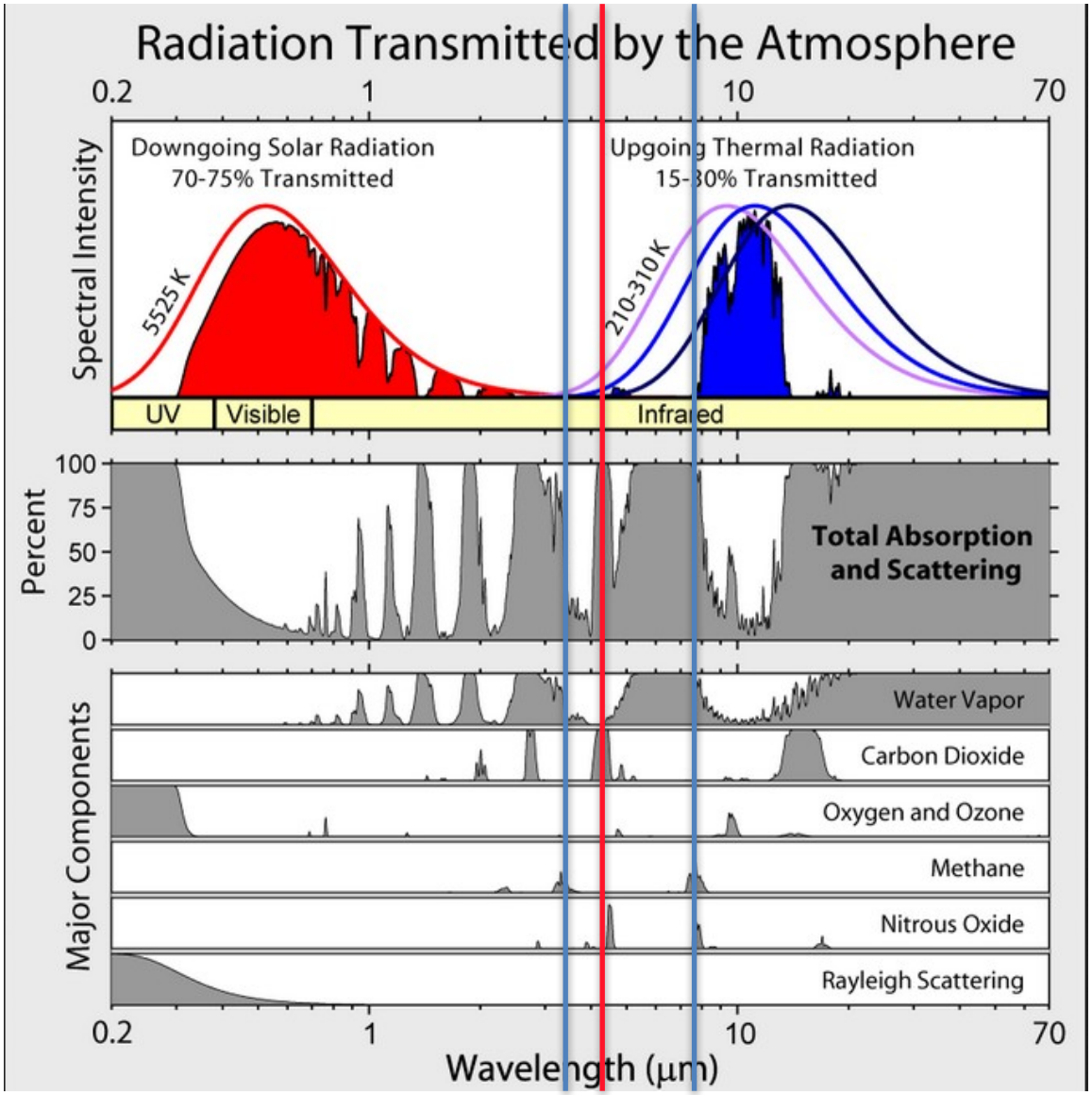
E_{input} = ?? = E_{output} = ??

$$T_E = (S_0 (1- \alpha) / (4 \sigma))^{1/4}$$

Greenhouse Effect



Greenhouse Effect

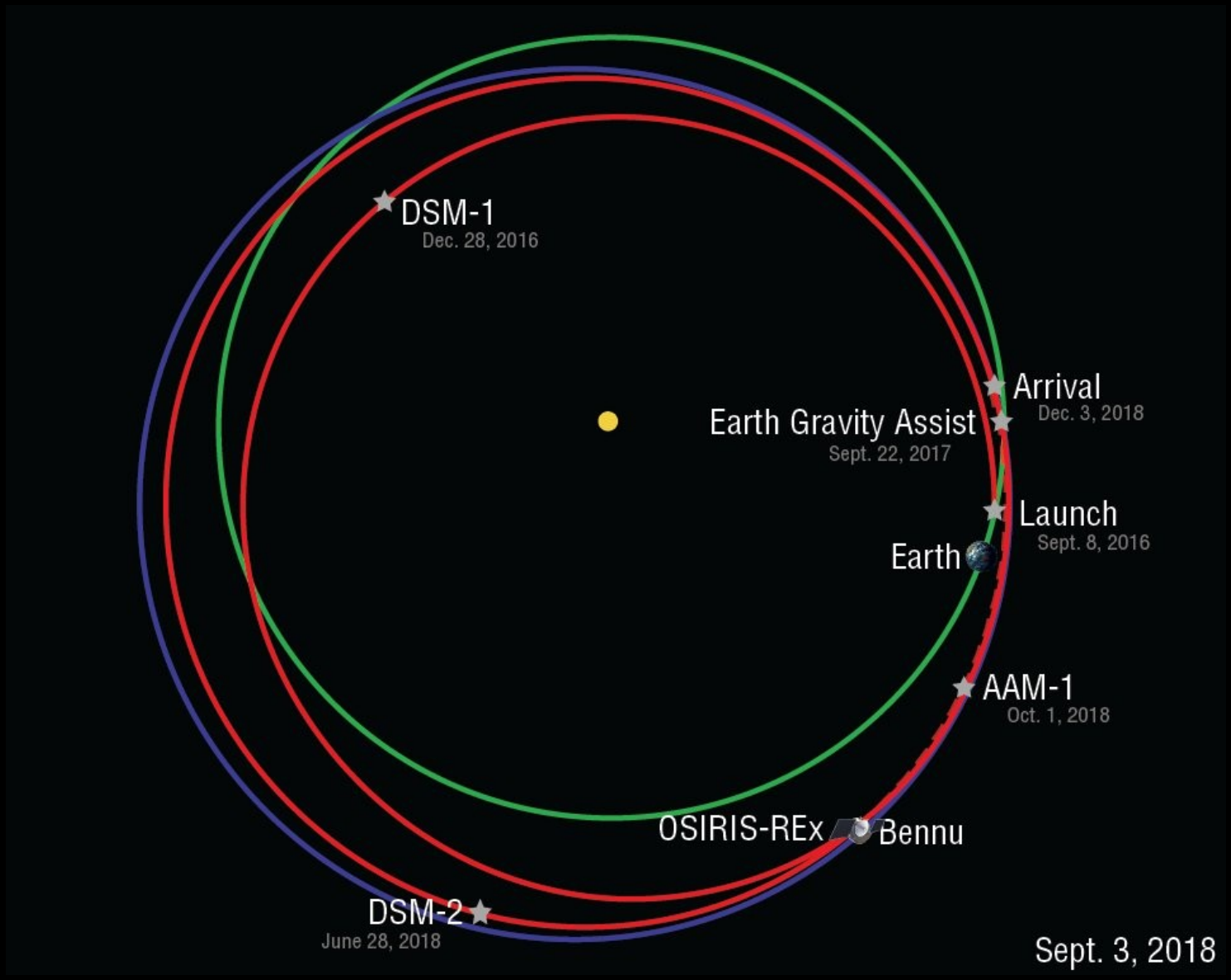


OSIRIS-REx



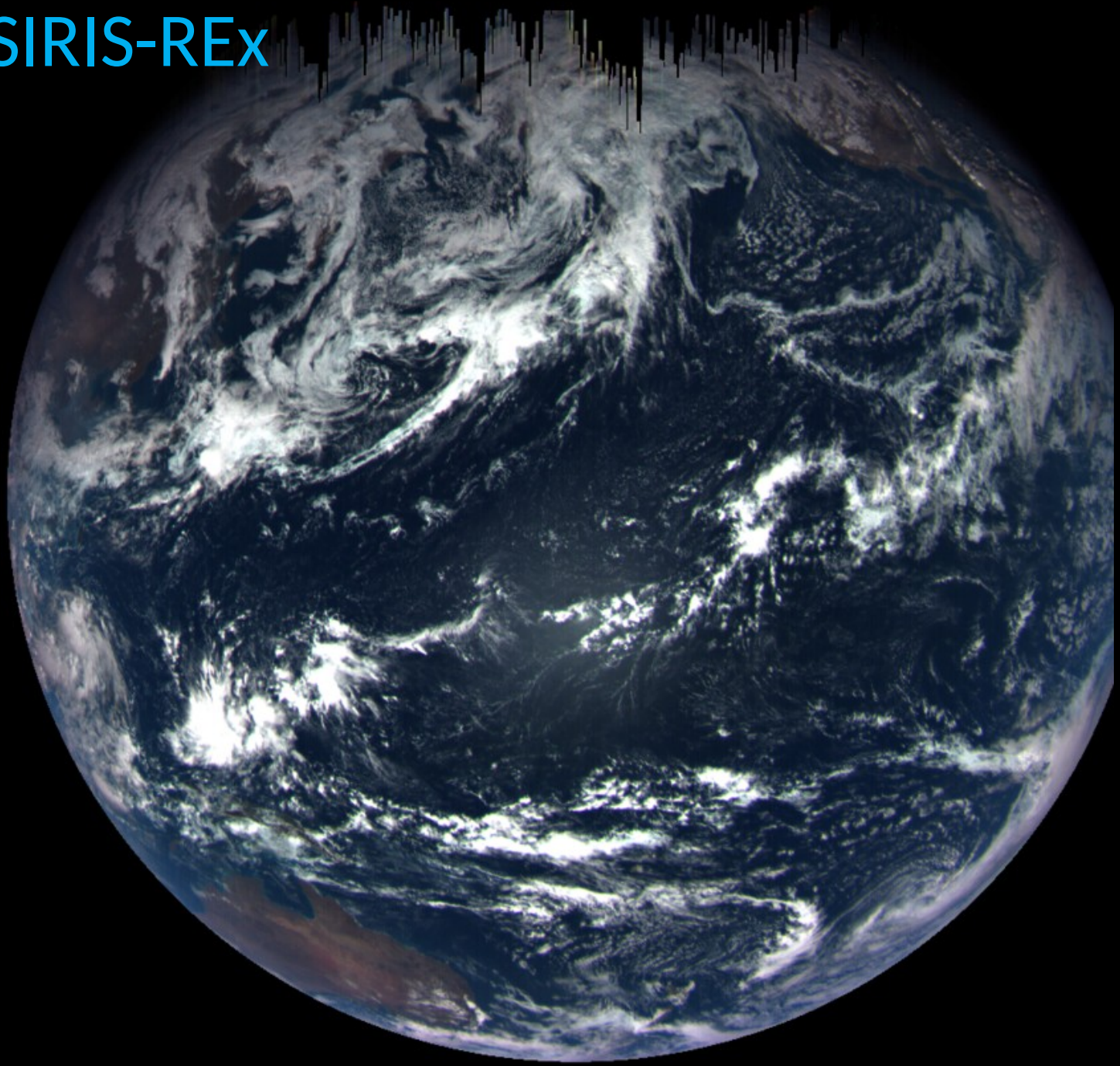
Sample return mission from an asteroid (101955 Bennu)

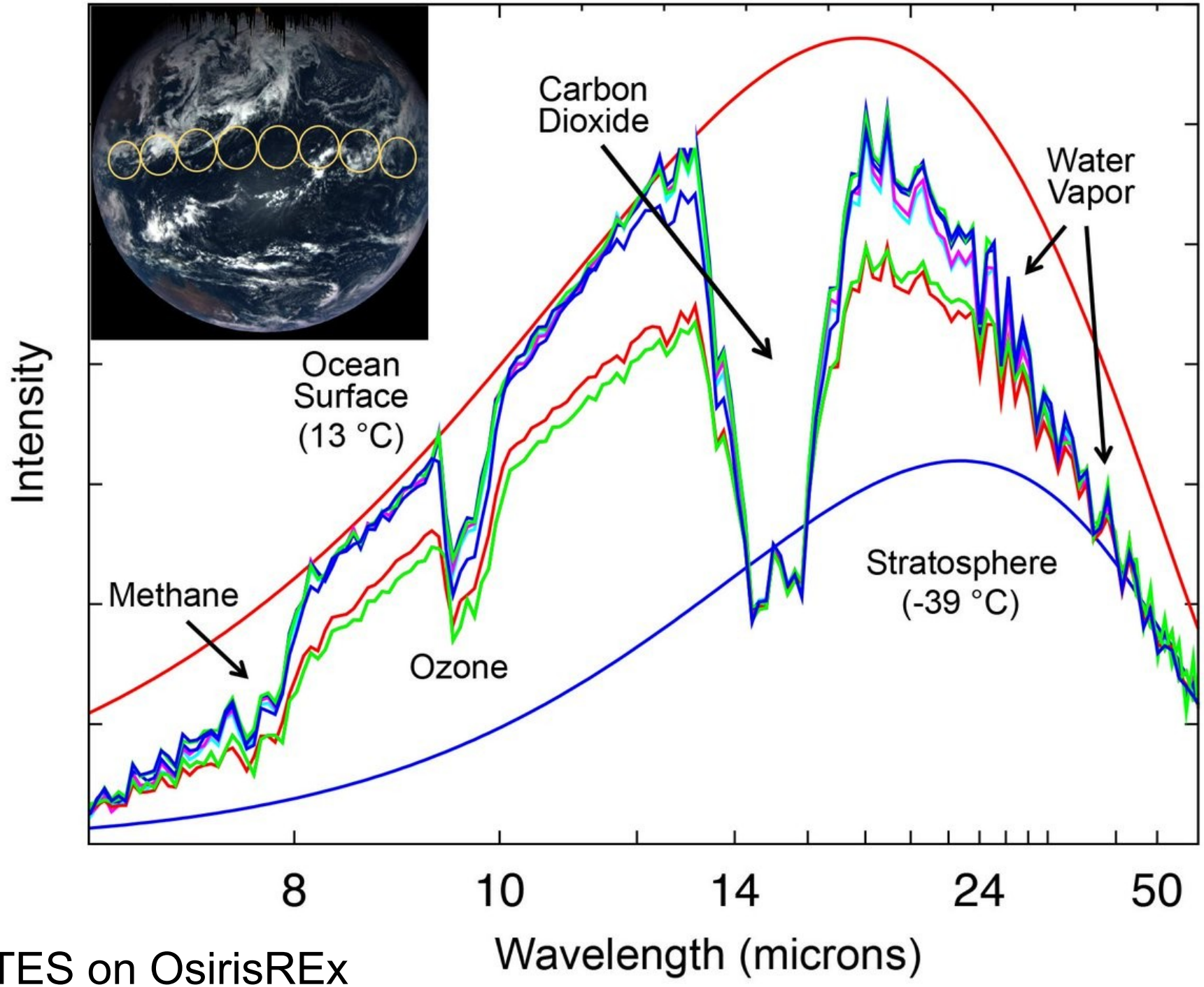
OSIRIS-REx



Sample return mission from an asteroid (101955 Bennu)

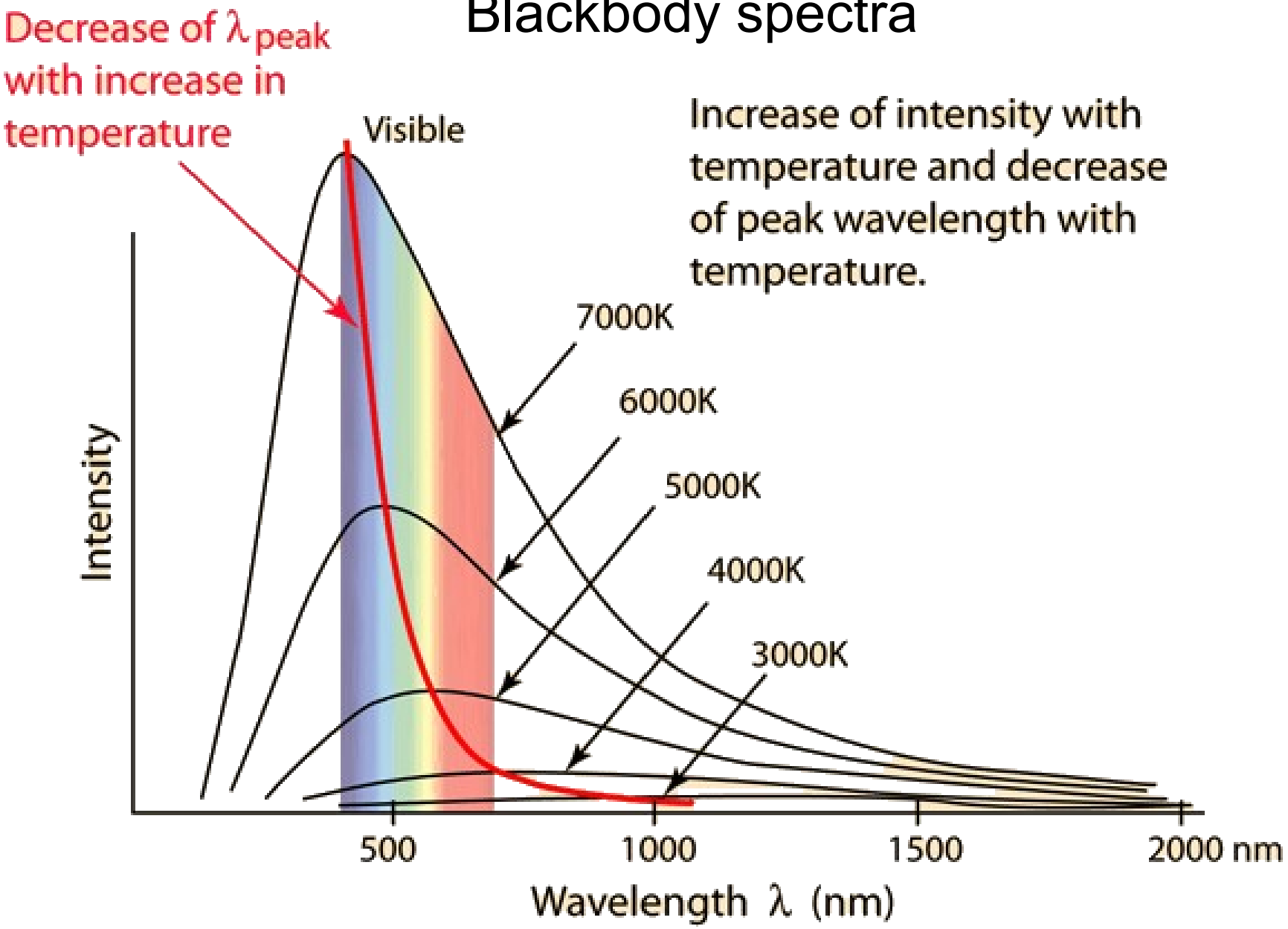
OSIRIS-REx





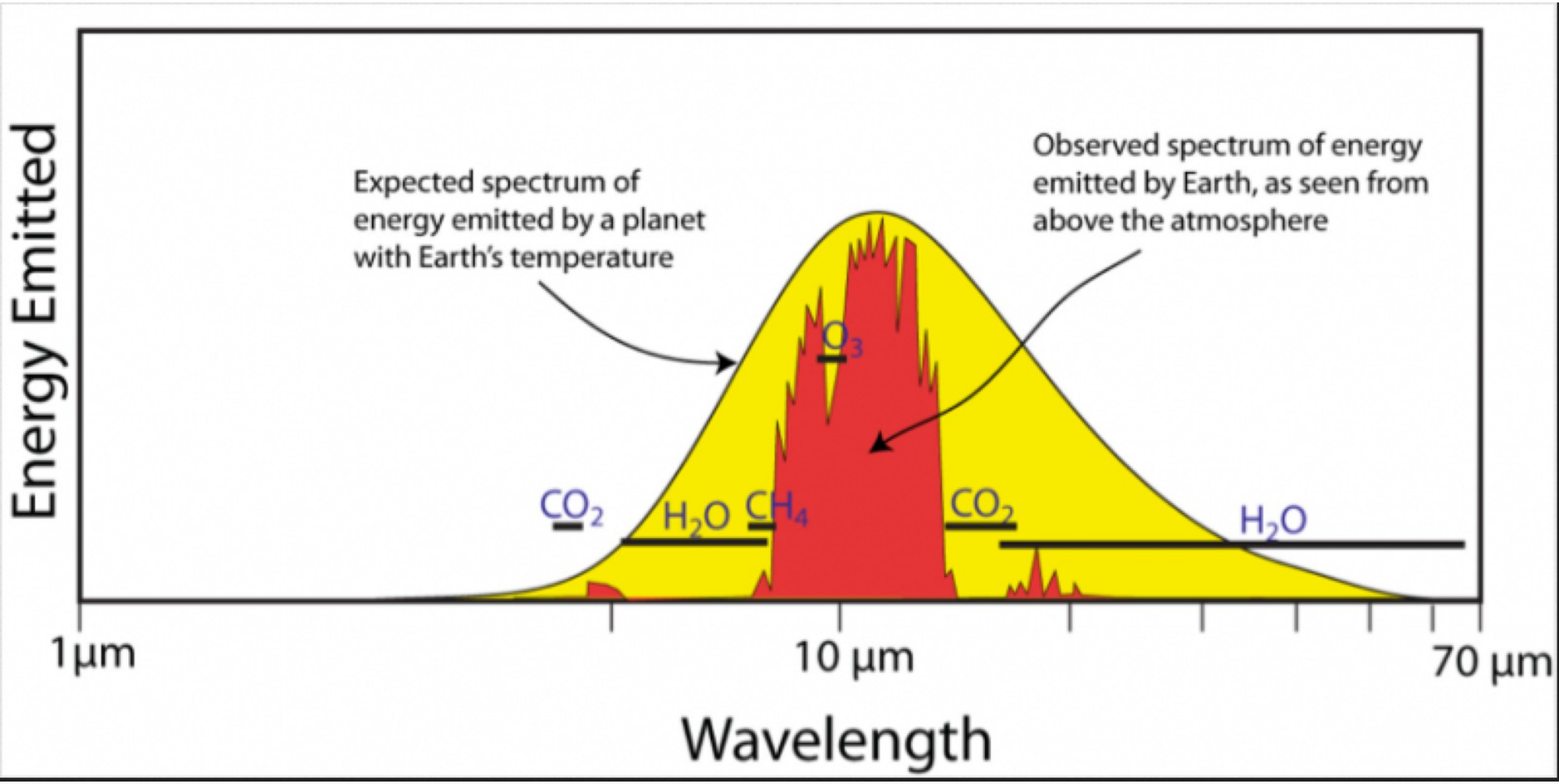
OTES on OsirisREx

Blackbody spectra

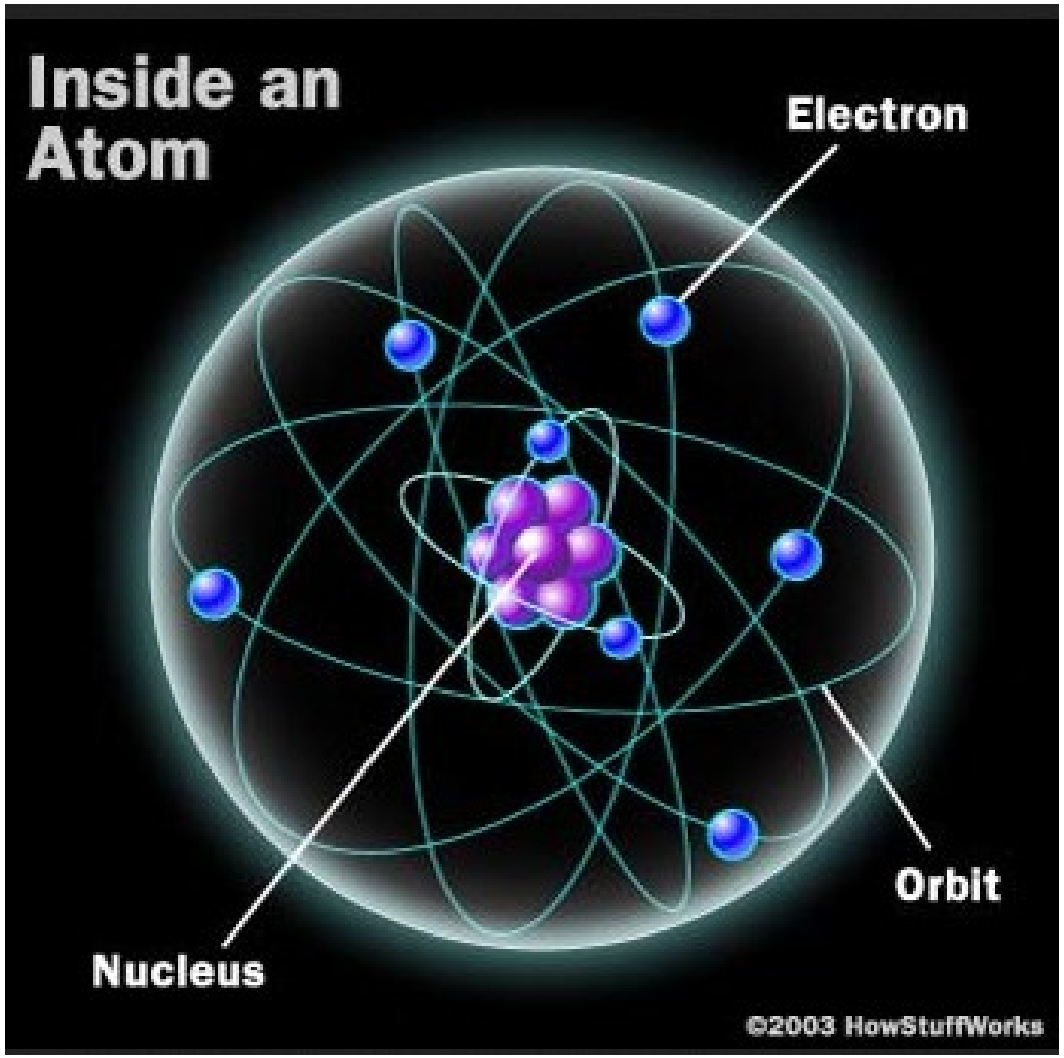


Blackbody emission scales as T^4

Spectrum of Earth Radiation
Top of the Atmosphere



Bohr's Model



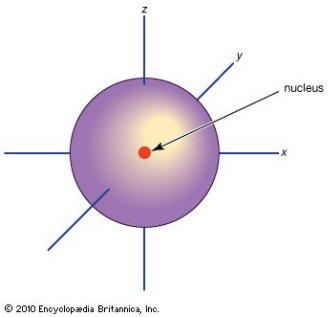
$$E = h f$$

E: Energy

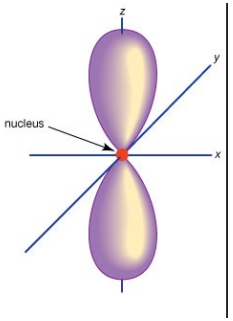
h: Plancks constant

f: Frequency

s orbital

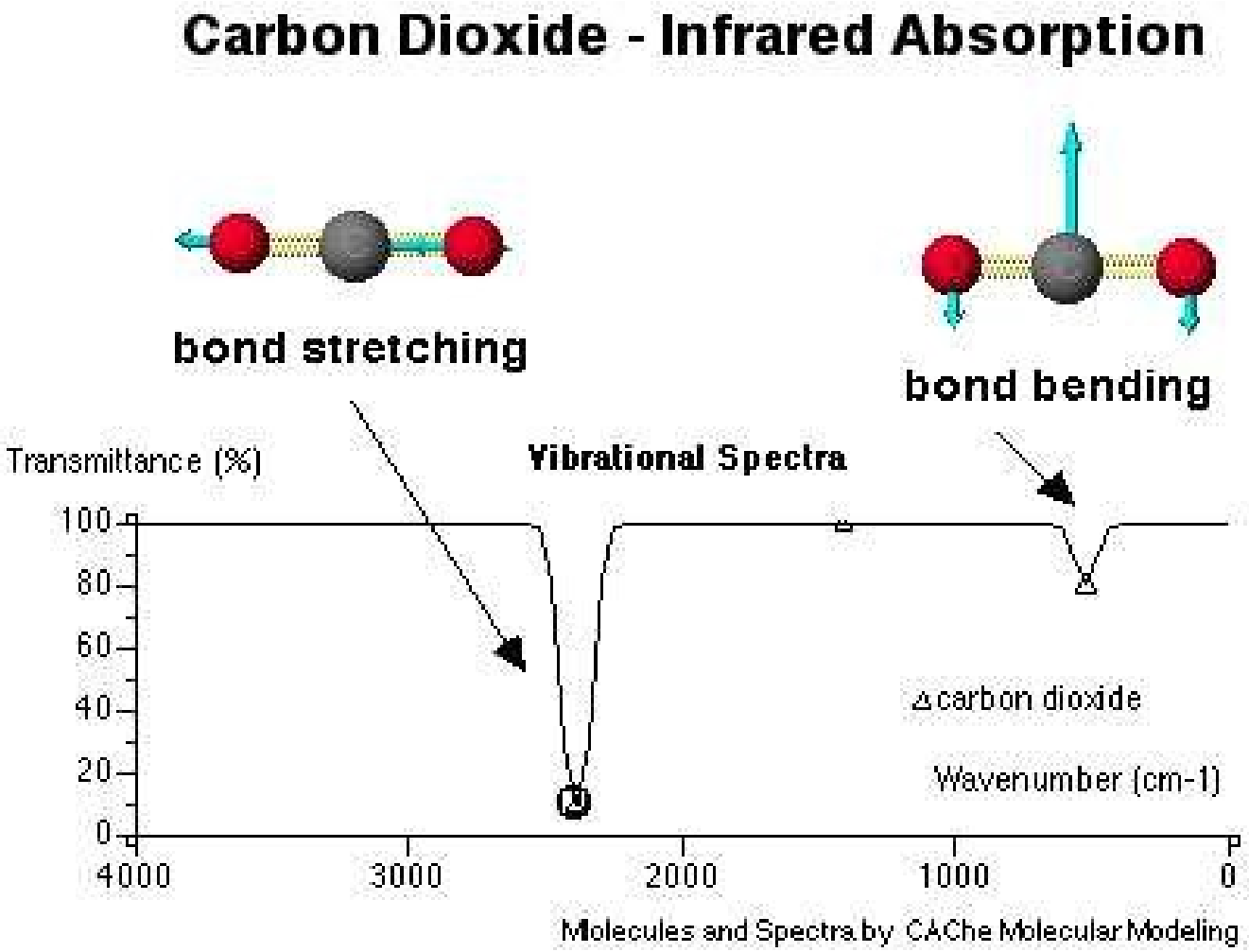


p orbital



Atoms and molecules can absorb only specific wavelength of frequency

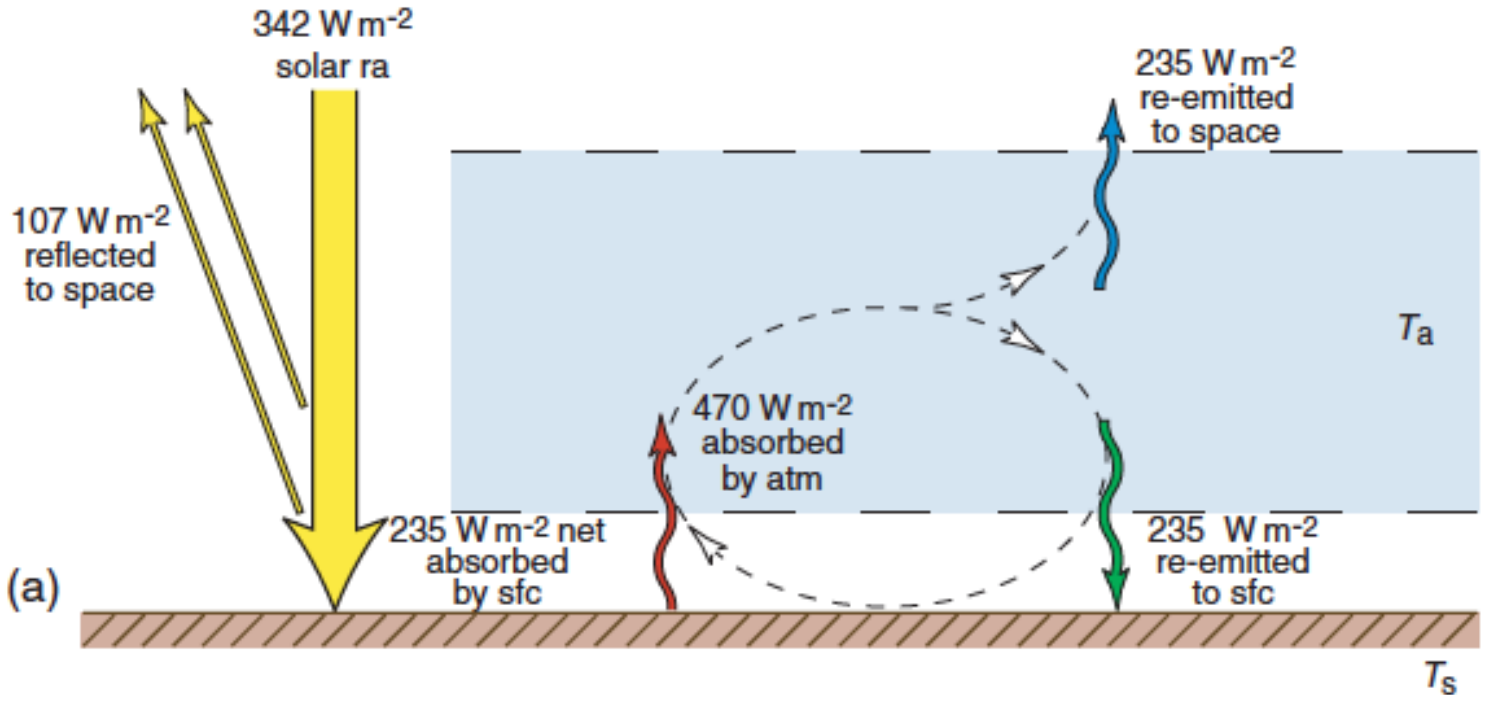
Infrared absorption: gasses



These absorb energy at distinct bands

Energy Balance
Greenhouse Effect

Black Body Atmosphere



Composition of the Atmosphere: San Francisco



August 2020: much of this is from forest fires; an estimated 1000 persons died prematurely just from this pollution alone.

One pollutant: PM2.5; this stands for particulate matter less than 2.5 microns in size

10 years ago: Composition of the Atmosphere: Beijing



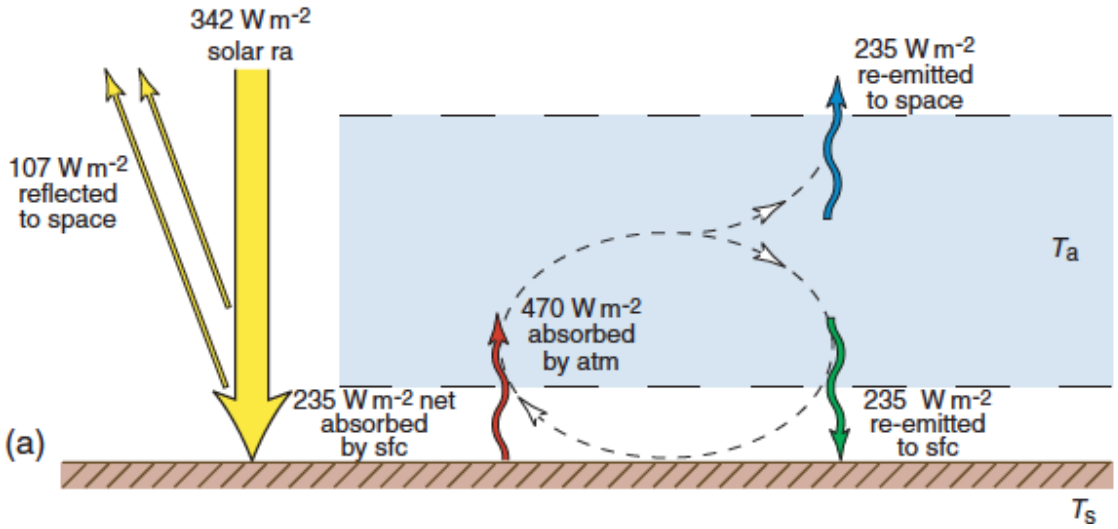
in Beijing: 568 micrograms per cubic metre (µg/m^3) of PM2.5 particles.

The WHO limit is 25 µg/m^3 that is: 25 times lower.

How much stuff is in the air, if we just consider the troposphere (10 km)?

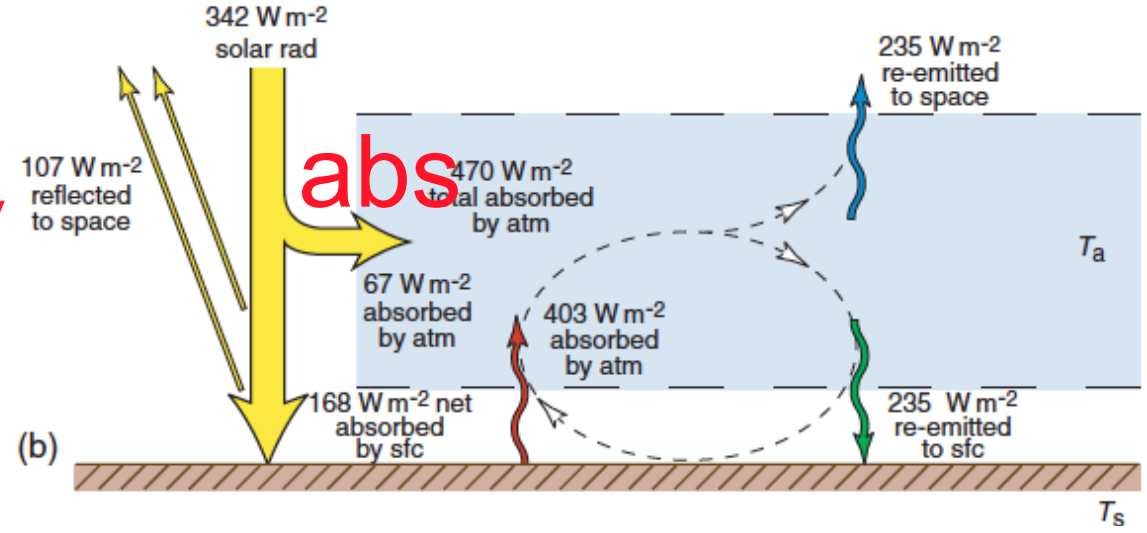
Energy Balance
Aerosol vs CO_2 forcing

Black Body
Atmosphere



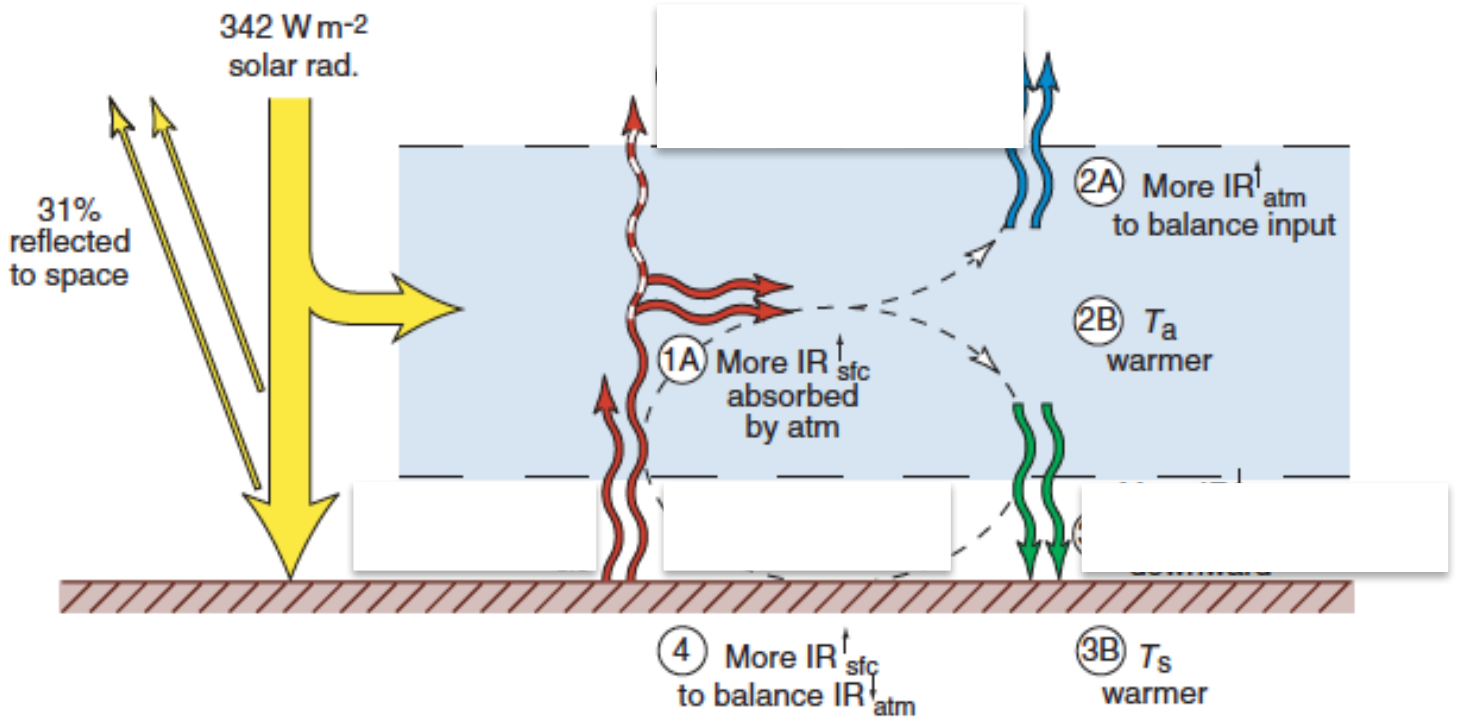
Global Dimming

abs = absorptivity

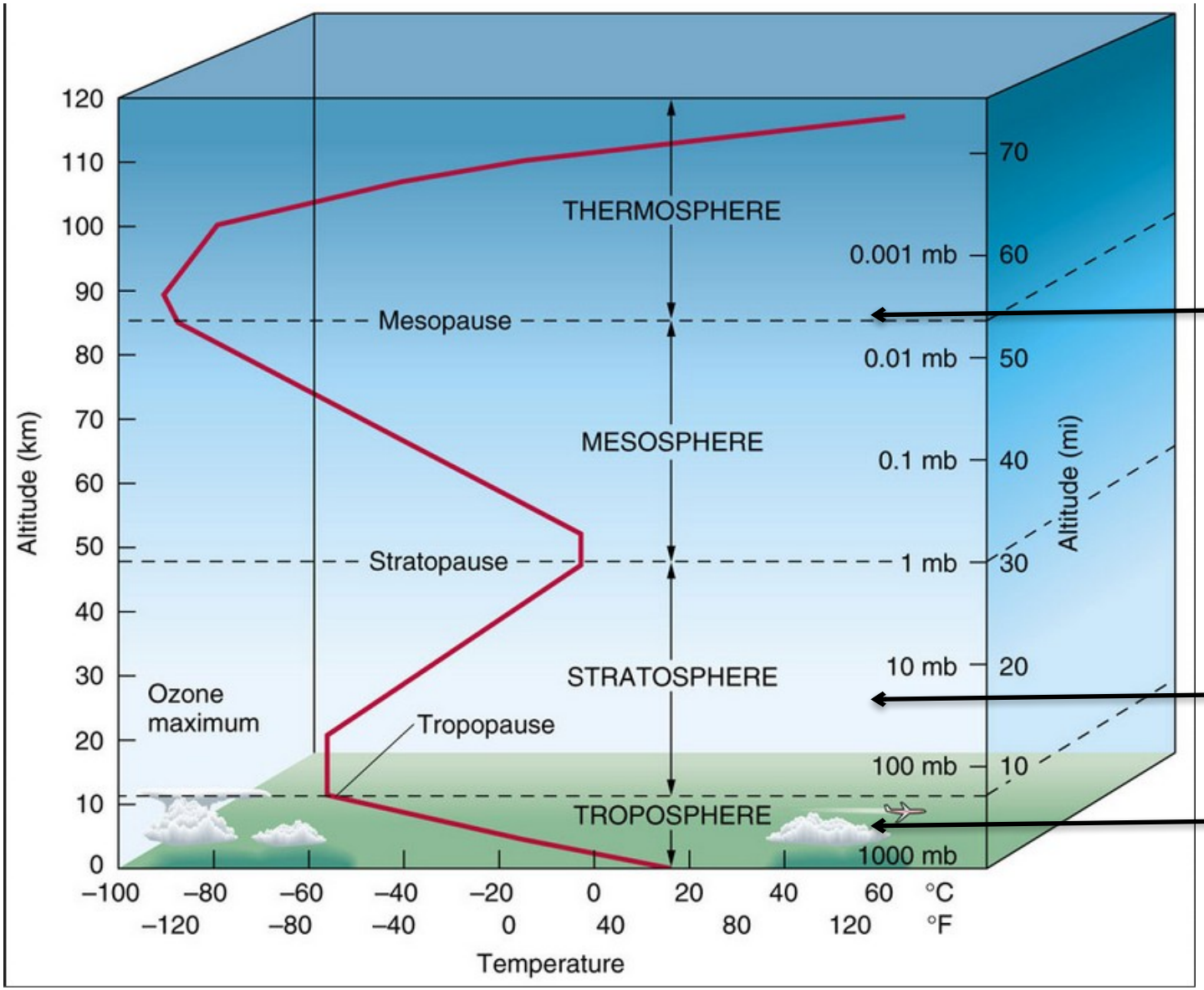


Energy Balance

Real Atmosphere – greenhouse effect



Vertical Structure of the Atmosphere

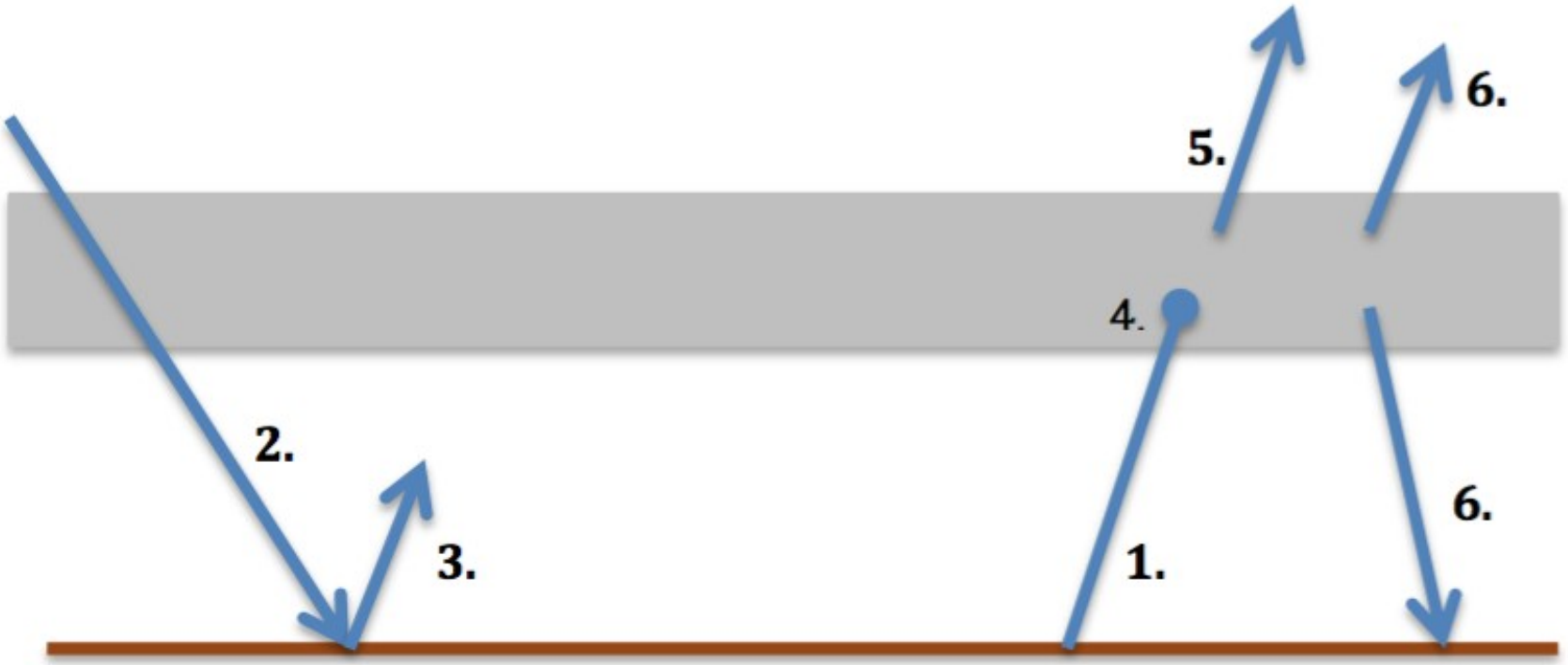


O_2 Aurora

O_3

Weather

Atmosphere Energy Balance



Atmosphere Energy Balance

Incoming solar shortwave radiation: $S_0/4$

Solar energy reflected back into space: the albedo α

Outgoing surface energy: σT_e^4

Surface longwave energy absorbed by the atmosphere: $\epsilon_a \sigma T_e^4$

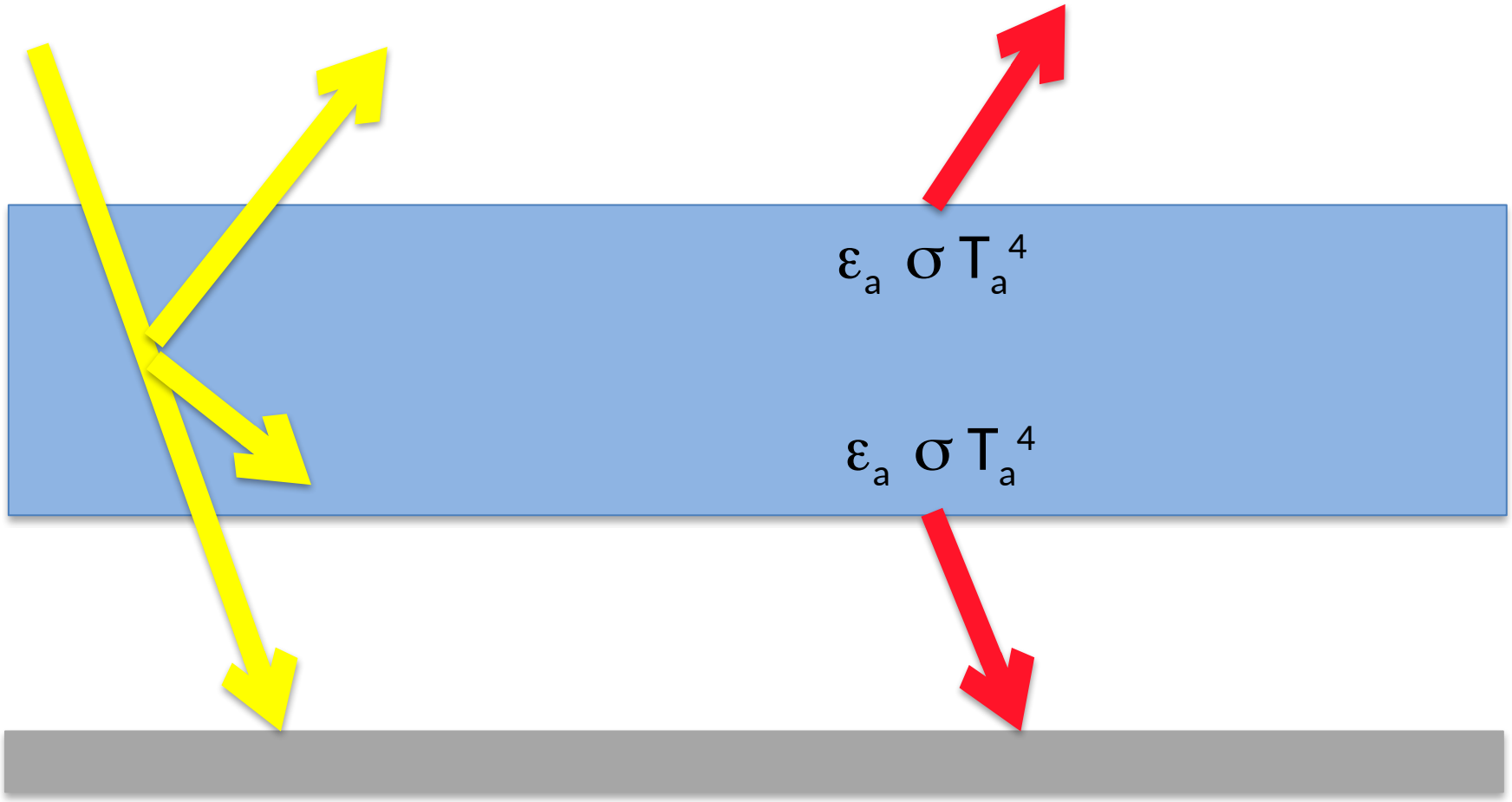
Rest of the surface longwave energy not absorbed by the atmosphere: $(1-\epsilon_a)\sigma T_e^4$

Atmospheric longwave energy emitted: $\epsilon_a \sigma T_a^4$

Earth's surface energy balance:

$$S_0/4(1-\alpha)+\epsilon_a \sigma T_a^4=\sigma T_e^4$$

Greenhouse Effect



OR?

